

Internship Report on Development of Vehicle-to-Infrastructure (V2I)- Based Traffic Signal Preemption for Emergency Vehicles

**Under the aegis of
DoT 5G Use Case Lab**



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Certificate

This is to certify that the internship report entitled "**Development of Vehicle-to-Infrastructure (V2I)-Based Traffic Signal Preemption for Emergency Vehicles**" has been prepared by **Mr. Piyush** and **Mr. Satyam Kumar**, students of the Department of Electronics and Communication Engineering, Delhi Technological University (DTU), as a partial fulfillment of the requirements of the **5G Use Case Internship Program**.

The work presented in this report was carried out at the **DoT 5G Use Case Lab, Delhi Technological University**, under proper guidance and supervision during the internship period. To the best of our knowledge, this report represents the original work carried out by the students and has not been submitted, either wholly or partly, for the award of any other degree, diploma, or certification.

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Declaration

We hereby declare that the internship report entitled "**Development of Vehicle-to-Infrastructure (V2I)-Based Traffic Signal Preemption for Emergency Vehicles**" is an original record of the work carried out by us during the **5G Use Case Internship Program** at the **DoT 5G Use Case Lab, Delhi Technological University**.

We further declare that this work has been completed under the guidance of **Dr. Ashish Kulkarni**, Assistant Professor, Department of Electrical Engineering, Delhi Technological University. The contents of this report have not been submitted previously, either in full or in part, for the award of any degree, diploma, or certificate at any other institution.

All references used in this report have been appropriately acknowledged.

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Abstract

The rapid growth of urban traffic congestion has significantly affected the response time of emergency vehicles, particularly ambulances that must pass through multiple signalized intersections while responding to critical situations. Conventional traffic signal systems operate using fixed timing plans and are unable to dynamically prioritize emergency vehicles, often resulting in unnecessary delays. The emergence of advanced vehicular communication technologies provides new opportunities to improve emergency traffic management through fast and reliable Vehicle-to-Infrastructure (V2I) communication.

This internship project presents the design, implementation, and performance evaluation of a Vehicle-to-Infrastructure (V2I) Traffic Signal Preemption System for Emergency Vehicles under three communication scenarios: **Baseline (No Communication)**, **4G LTE Cloud-Routed Preemption**, and **5G URLLC Direct PC5 Sidelink Preemption**. The proposed system was developed using the **Simulation of Urban MObility (SUMO)** traffic simulator integrated with **Python Traffic Control Interface (TraCI)** for real-time traffic signal control and a **Streamlit dashboard** for visualization and performance analysis. All scenarios were tested under identical traffic conditions to ensure a fair comparison of system behavior and performance.

The evaluation focuses on key performance parameters, including emergency vehicle travel time, average speed, stopped delay, and traffic signal response. Simulation results show that the Baseline scenario experiences the highest delay due to conventional fixed-time traffic signal operation. The 4G LTE communication model improves emergency vehicle movement by enabling traffic signal preemption; however, its communication latency and limited reaction window reduce the effectiveness of signal transition. In contrast, the 5G URLLC communication model, with ultra-low latency and earlier communication initiation, enables more timely traffic signal preemption and allows the emergency vehicle to pass through intersections with minimal stopping. The study further highlights that communication range, latency, and traffic signal safety clearance are all important factors in achieving efficient emergency traffic management.

The proposed implementation demonstrates the effectiveness of comparing baseline, 4G LTE, and 5G URLLC communication approaches for intelligent traffic signal control and provides a scalable simulation framework for future research in connected vehicles, Vehicle-to-Infrastructure communication, and intelligent transportation systems. The internship offered valuable practical experience in traffic simulation, communication system modelling, and software-based implementation of emerging vehicular communication use cases.

v
Contents

Certificate	i
Declaration	ii
Acknowledgement.....	iii
Abstract.....	iv
List of Abbreviations	viii
Chapter 1	1
Introduction	1
1.1 Background of Mobile Communication Generations (1G–5G).....	2
1.2 Motivation for 5G.....	6
1.3 Applications of 5G	7
1.4 Challenges Addressed by 5G	8
1.5 Relevance of 5G to Emergency Vehicle Preemption	8
1.6 Chapter Summary.....	9
Chapter 2	10
Key Features of 5G.....	10
2.1 Enhanced Mobile Broadband (eMBB).....	10
2.2 Ultra-Reliable Low-Latency Communication (URLLC).....	18
2.3 Massive Machine-Type Communication (mMTC).....	29
2.4 Overall Chapter Summary.....	37
Chapter 3	39
5G Network Architecture	39
3.1 Overview of 5G Architecture	39
3.2 User Equipment (UE).....	40
3.3 5G Radio Access Network (RAN)	40
3.4 gNodeB (gNB)	41
3.5 5G Core Network (5GC)	41
3.6 Major Functional Components of the 5G Core.....	42
3.7 Service-Based Architecture (SBA)	43
3.8 Network Slicing.....	44
3.9 Multi-Access Edge Computing (MEC).....	45
3.10 Network Function Virtualization (NFV)	45

3.11 Software Defined Networking (SDN).....	45
3.12 Security in 5G Networks.....	46
3.13 Integration with Intelligent Transportation Systems.....	46
3.14 Relevance to the Internship Project.....	47
3.15 Chapter Summary.....	48
Chapter 4	49
5G Standards	49
4.1 Role of Standardization in Mobile Communication	49
4.2 Third Generation Partnership Project (3GPP).....	50
4.3 ITU IMT-2020 Framework	53
4.4 5G New Radio (NR).....	54
4.5 Frequency Spectrum.....	55
4.6 Quality of Service (QoS).....	56
4.7 Security Standards.....	56
4.8 Standards Supporting Intelligent Transportation Systems.....	57
4.9 Relevance to the Internship Project.....	57
4.10 Chapter Summary.....	58
Chapter 5	59
Development of Vehicle-to-Infrastructure (V2I)-Based Traffic Signal Preemption for Emergency Vehicles	59
5.1 Introduction	59
5.2 Theory and Network Modeling.....	60
5.3 Implementation Details	65
5.4 Results and Analysis.....	72
Chapter 6	85
Conclusion and Future Scope.....	85
6.1 Conclusion.....	85
6.2 Future Scope.....	86
6.3 Internship Learning Outcomes.....	87
6.4 Final Remarks.....	88
References	89

Figures

Figure 1.1: Evolution of Mobile Communication from 1G to 5G (timeline diagram).....	6
Figure 2.1: Massive MIMO Antenna System	12
Figure 2.2: Beamforming Concept.....	13
Figure 2.3: Small Cell Deployment.....	14
Figure 2.4: Edge Computing Architecture	21
Figure 2.5: Network Slicing Concept.....	22
Figure 2.6: C-V2X Communication Architecture.....	27
Figure 3.1: Overall 5G Network Architecture.....	39
Figure 3.2: Service-Based Architecture (SBA) of the 5G Core.....	44
Figure 3.3: Communication Flow for Emergency Vehicle Preemption.....	47
Figure 4.1: Timeline of 3GPP Releases.....	51
Figure 5.1: 4G LTE Cloud Routing Path showing gNB, Core, Cloud Server, and RSU Interfaces	62
Figure 5.2: 5G PC5 Sidelink Direct Communication Path bypassing Base Station Routing.	63
Figure 5.3: Traffic Light Phase Sequence Transition Diagram (NS Green to Yellow to EW Green)	64
Figure 5.4: Overall System Architecture.....	66
Figure 5.5: SUMO Node and Edge XML Code Snippet and Attribute Configurations	67
Figure 5.6: SUMO Road Network	68
Figure 5.7: Traffic Signal Preemption Flowchart	71
Figure 5.8: Baseline Simulation Output.....	75
Figure 5.9: 4G LTE Preemption Scenario	76
Figure 5.10: 5G URLLC Preemption Scenario.....	78
Figure 5.11: Latency Histogram Comparison showing 4G Retransmission Spikes and 5G Stable Profiles.....	79
Figure 5.12: Streamlit Dashboard Layout showing Sidebars and Results Visualization Panels	80
Figure 5.13: Plotly Chart showing speed profile of the Ambulance under the Three Modes	82
Figure 5.14: Comparison Bar Chart of Total Travel Time Across All Modes	82
Figure 5.15: Comparison Bar Chart of Stopped Delay Across All Modes	83

List of Abbreviations

Abbreviation	Full Form
3GPP	Third Generation Partnership Project
AI	Artificial Intelligence
AMF	Access and Mobility Management Function
C-ITS	Cooperative Intelligent Transportation Systems
C-V2X	Cellular Vehicle-to-Everything
DTU	Delhi Technological University
EV	Emergency Vehicle
gNB	Next Generation NodeB
ITS	Intelligent Transportation System
LTE	Long Term Evolution
MEC	Multi-access Edge Computing
NR	New Radio
PC5	Proximity Communication Interface
QoS	Quality of Service
RSU	Road Side Unit
SBA	Service-Based Architecture
SUMO	Simulation of Urban MObility
TraCI	Traffic Control Interface
UE	User Equipment
URLLC	Ultra-Reliable Low-Latency Communication
V2I	Vehicle-to-Infrastructure
V2V	Vehicle-to-Vehicle

Chapter 1

Introduction

The evolution of wireless communication has transformed the way people communicate, access information, and interact with digital technologies. From simple analog voice communication in the early cellular generations to today's intelligent, high-speed, and highly reliable communication networks, mobile technology has continuously evolved to meet increasing user demands. The emergence of the Internet of Things (IoT), autonomous vehicles, smart cities, industrial automation, cloud computing, artificial intelligence, and immersive multimedia applications has significantly increased the requirements for higher data rates, lower latency, greater reliability, and massive connectivity.

The fifth generation (5G) of mobile communication represents a revolutionary advancement over previous generations by providing a unified communication platform capable of supporting enhanced mobile broadband (eMBB), ultra-reliable low-latency communication (URLLC), and massive machine-type communication (mMTC). Unlike previous generations that primarily focused on improving data speed, 5G has been designed to support a diverse range of applications requiring real-time communication, intelligent decision-making, and seamless connectivity among billions of devices.

One of the most promising applications of 5G is its integration with Intelligent Transportation Systems (ITS). Road traffic congestion remains one of the major challenges in urban environments, often delaying emergency vehicles such as ambulances, fire engines, and police vehicles. Even a few seconds of delay during emergency response can have serious consequences for public safety and healthcare outcomes. Traditional traffic signal systems operate using fixed timing plans and are unable to dynamically prioritize emergency vehicles based on real-time conditions.

The ultra-low latency and ultra-high reliability provided by 5G URLLC enable emergency vehicles to communicate directly with roadside infrastructure and traffic signal controllers. This allows intelligent traffic signal preemption, where intersections automatically provide a green signal to approaching emergency vehicles, thereby minimizing delays and improving response times.

This internship focuses on the Development of Vehicle-to-Infrastructure (V2I)-Based Traffic Signal Preemption for Emergency Vehicles using the SUMO traffic simulator integrated with Python TraCI. The study compares the performance of conventional traffic control, 4G LTE-based preemption, and 5G URLLC-based preemption under realistic urban traffic scenarios. Through simulation and performance analysis, the project demonstrates how next-generation wireless communication can improve transportation efficiency while maintaining acceptable traffic conditions for civilian vehicles.

1.1 Background of Mobile Communication Generations (1G–5G)

Mobile communication has undergone remarkable evolution over the past four decades. Each generation has introduced significant technological advancements to overcome the limitations of its predecessor while addressing emerging communication requirements.

First Generation (1G)

The first generation of mobile communication was introduced during the early 1980s and relied entirely on analog transmission techniques. It primarily supported voice communication using Frequency Division Multiple Access (FDMA). Although revolutionary for its time, 1G suffered from poor voice quality, limited coverage, low spectral efficiency, and inadequate security due to the absence of encryption.

Characteristics

- Analog communication
- Voice-only services
- FDMA technology
- Low capacity
- Poor security
- Limited coverage

Limitations

- Poor voice quality
- High interference
- No data transmission
- No encryption
- Limited number of users

Second Generation (2G)

The introduction of 2G marked the transition from analog to digital communication. Technologies such as GSM and CDMA significantly improved voice quality while introducing SMS messaging and basic data services. Digital modulation techniques also improved security through encryption.

Characteristics

- Digital communication
- GSM and CDMA
- SMS support
- Basic Internet access
- Improved security

Advantages

- Better voice quality
- Increased capacity
- Lower power consumption
- Digital encryption
- International roaming

Limitations

- Very low data rates
 - Limited multimedia capability
 - Not suitable for broadband applications
-

Third Generation (3G)

Third-generation networks were designed to support mobile Internet and multimedia communication. Technologies such as UMTS and WCDMA enabled higher data rates and introduced services including video calling, web browsing, GPS navigation, and multimedia streaming.

Characteristics

- Mobile Internet
- Video calling
- Multimedia support
- Packet-switched data
- Improved mobility

Advantages

- Faster Internet access

- Better multimedia experience
- Simultaneous voice and data communication

Limitations

- Higher infrastructure cost
 - Limited spectrum efficiency
 - Increased network complexity
-

Fourth Generation (4G)

Fourth-generation communication introduced all-IP networking through LTE and LTE-Advanced technologies. 4G significantly increased network speed, reduced latency, and enabled seamless multimedia services.

Characteristics

- LTE technology
- All-IP architecture
- High-definition video streaming
- Cloud applications
- Voice over LTE (VoLTE)

Advantages

- High-speed Internet
- Lower latency
- Better spectral efficiency
- Improved Quality of Service

Limitations

Although 4G provides excellent broadband performance, it cannot consistently satisfy applications requiring ultra-low latency and extremely high reliability, such as autonomous driving, industrial automation, and emergency vehicle communication.

Fifth Generation (5G)

The fifth generation represents a paradigm shift in wireless communication by extending beyond faster Internet connectivity. 5G has been designed to support intelligent and mission-critical services through three major service categories:

- Enhanced Mobile Broadband (eMBB)
- Ultra-Reliable Low-Latency Communication (URLLC)
- Massive Machine-Type Communication (mMTC)

Unlike previous generations, 5G enables communication between humans, machines, vehicles, infrastructure, and cloud services within a unified ecosystem.

Major Features

- Multi-gigabit data rates
- Ultra-low latency (1–5 ms)
- Massive IoT connectivity
- Network slicing
- Edge computing integration
- Artificial Intelligence support
- Ultra-high reliability (99.999%)

Comparison of Mobile Generations

<i>Generation</i>	<i>Main Service</i>	<i>Data Rate</i>	<i>Typical Latency</i>	<i>Major Technology</i>
1G	Analog Voice	Up to 2.4 kbps	High	FDMA
2G	Digital Voice & SMS	Up to 64 kbps	~400 ms	GSM, CDMA
3G	Mobile Internet	Up to 2 Mbps	~200 ms	UMTS, WCDMA
4G	Broadband Internet	Up to 1 Gbps	10–50 ms	LTE
5G	Intelligent Connectivity	Up to 20 Gbps	1–5 ms	NR, Massive MIMO, Beamforming

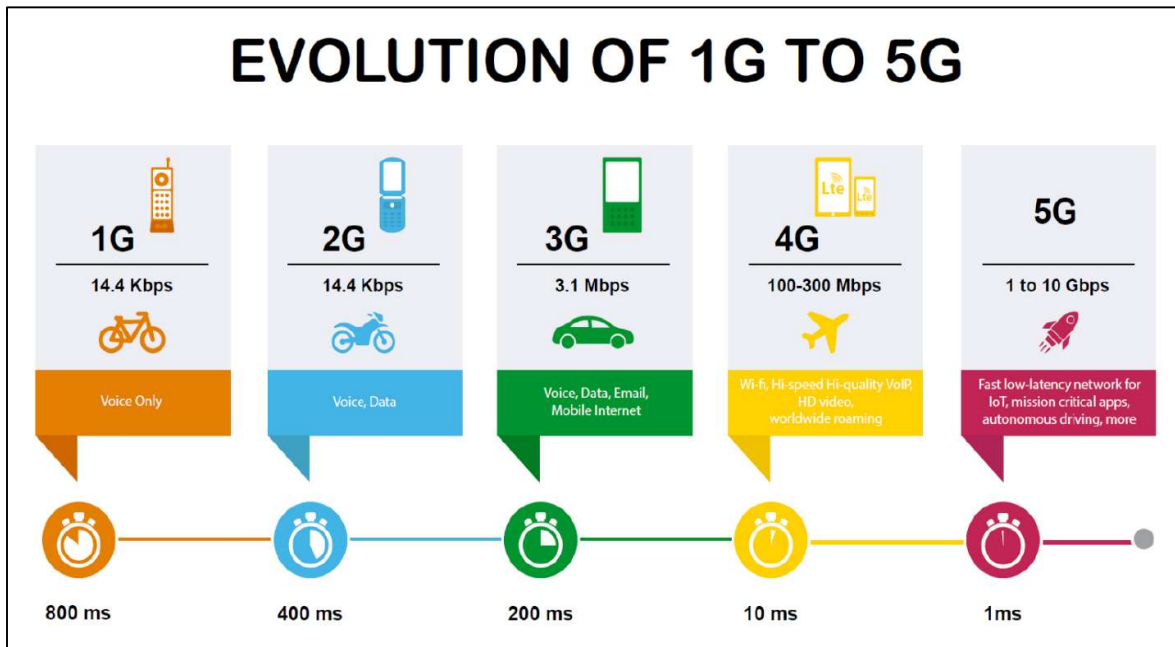


Figure 1.1: Evolution of Mobile Communication from 1G to 5G (timeline diagram).

1.2 Motivation for 5G

The rapid increase in connected devices, cloud services, and intelligent applications has exposed the limitations of existing communication networks. Applications such as autonomous vehicles, industrial robotics, remote healthcare, augmented reality, smart grids, and emergency response systems require communication networks capable of delivering extremely low latency, high reliability, and massive connectivity.

Several factors motivated the development of 5G:

- Exponential growth in mobile data traffic.
- Increasing demand for ultra-high-definition multimedia.
- Large-scale deployment of IoT devices.
- Growth of Industry 4.0 and smart manufacturing.
- Development of connected and autonomous vehicles.
- Requirement for mission-critical communication.
- Expansion of smart city infrastructure.
- Need for energy-efficient and scalable communication networks.

5G was therefore developed not only to increase network speed but also to create a flexible communication platform capable of supporting diverse application domains.

1.3 Applications of 5G

The capabilities of 5G extend across numerous industries.

Healthcare

- Remote surgery
- Telemedicine
- Patient monitoring
- Connected medical devices

Transportation

- Intelligent transportation systems
- Vehicle-to-Everything (V2X)
- Emergency vehicle preemption
- Autonomous vehicles

Smart Cities

- Intelligent traffic management
- Smart lighting
- Environmental monitoring
- Smart surveillance

Industrial Automation

- Robotics
- Predictive maintenance
- Factory automation
- Digital twins

Agriculture

- Precision farming
- Smart irrigation
- Livestock monitoring
- Autonomous agricultural machinery

Entertainment

- Cloud gaming

- Virtual reality
 - Augmented reality
 - 8K video streaming
-

1.4 Challenges Addressed by 5G

Although previous mobile generations successfully improved communication capabilities, several challenges remained unresolved.

These include:

- High communication latency.
- Network congestion.
- Limited support for IoT devices.
- Poor reliability for mission-critical applications.
- Limited scalability.
- Inadequate spectrum utilization.
- Lack of intelligent network management.

5G introduces technologies such as Network Slicing, Software Defined Networking (SDN), Network Function Virtualization (NFV), Edge Computing, Massive MIMO, Beamforming, and Artificial Intelligence to overcome these challenges.

1.5 Relevance of 5G to Emergency Vehicle Preemption

One of the most impactful applications of URLLC is emergency vehicle traffic management. Conventional traffic signals operate independently and cannot react quickly to emergency vehicle movement. Consequently, ambulances and fire engines often experience delays caused by traffic congestion.

Using 5G-enabled Cellular Vehicle-to-Everything (C-V2X) communication, emergency vehicles can continuously exchange information with roadside infrastructure. The traffic controller receives the vehicle's location, speed, and direction in real time and dynamically changes signal phases before the vehicle reaches the intersection.

Compared to 4G LTE, the significantly lower communication latency of 5G allows earlier traffic signal activation, enabling downstream traffic queues to clear more efficiently. This improves emergency response times while minimizing disruption to normal traffic.

The internship project presented in this report investigates this concept through simulation using SUMO, Python TraCI, and Streamlit by comparing three scenarios: Baseline (No Preemption), 4G LTE Preemption, and 5G URLLC Preemption.

1.6 Chapter Summary

This chapter introduced the evolution of mobile communication technologies from 1G to 5G and discussed the motivation behind developing next-generation wireless networks. The major capabilities of 5G, including enhanced broadband, ultra-low latency, and massive device connectivity, were presented along with their applications across healthcare, transportation, industrial automation, and smart cities. Finally, the chapter established the relevance of 5G to intelligent transportation systems and emergency vehicle preemption, providing the foundation for the theoretical concepts discussed in the subsequent chapters.

Chapter 2

Key Features of 5G

The fifth generation (5G) of mobile communication has been designed to address the diverse communication requirements of modern digital applications. Unlike previous generations that primarily focused on improving voice quality or increasing data rates, 5G introduces a flexible and service-oriented architecture capable of supporting a wide variety of use cases. To achieve this objective, the International Telecommunication Union (ITU) and the 3rd Generation Partnership Project (3GPP) categorize 5G services into three major communication scenarios:

- **Enhanced Mobile Broadband (eMBB)**
- **Ultra-Reliable Low-Latency Communication (URLLC)**
- **Massive Machine-Type Communication (mMTC)**

These three service categories collectively enable applications ranging from ultra-high-speed multimedia streaming to autonomous vehicles, industrial automation, smart cities, healthcare, and billions of connected IoT devices.

This chapter discusses each of these key features in detail.

2.1 Enhanced Mobile Broadband (eMBB)

2.1.1 Introduction

Enhanced Mobile Broadband (eMBB) is one of the primary service categories introduced in 5G networks. Its primary objective is to provide significantly higher data rates, improved network capacity, wider coverage, and a better user experience compared to previous mobile communication generations.

With the increasing popularity of cloud computing, video streaming, virtual reality, augmented reality, and online gaming, existing communication networks often struggle to meet user demands during peak traffic conditions. eMBB addresses these challenges by utilizing advanced radio technologies, wider spectrum bandwidth, and intelligent resource allocation techniques to deliver multi-gigabit data rates with consistent performance.

The deployment of eMBB enables users to access high-quality digital services regardless of whether they are in densely populated urban areas, transportation hubs, or rural environments.

2.1.2 Objectives of eMBB

The major objectives of Enhanced Mobile Broadband include:

- Deliver extremely high download and upload speeds.
 - Improve user experience in crowded environments.
 - Support ultra-high-definition multimedia services.
 - Increase network capacity for growing mobile traffic.
 - Enhance spectral efficiency.
 - Provide seamless mobility across different coverage areas.
 - Enable new digital services requiring high bandwidth.
-

2.1.3 Key Characteristics

The major characteristics of eMBB include:

High Data Rates

5G eMBB supports theoretical peak download speeds of up to **20 Gbps** and upload speeds approaching **10 Gbps**, depending on spectrum allocation and deployment scenarios.

Large Bandwidth

Unlike previous generations, 5G utilizes much wider bandwidth channels, particularly in the millimeter-wave (mmWave) spectrum. Wider bandwidth directly increases network throughput and supports multiple simultaneous high-speed users.

Improved Spectral Efficiency

Advanced technologies such as Massive MIMO and Beamforming allow more efficient utilization of available radio spectrum, resulting in higher network capacity without requiring proportionally more spectrum resources.

High User Capacity

Modern urban environments often contain thousands of connected users within a small geographical area. eMBB is designed to maintain high performance even in crowded locations such as:

- Stadiums
- Railway stations
- Airports

- Shopping malls
- Concert venues
- Smart campuses

Seamless Mobility

Users moving at high speeds, such as passengers on high-speed trains or vehicles, can continue receiving stable broadband connectivity with minimal interruption.

2.1.4 Technologies Enabling eMBB

Several advanced technologies work together to achieve the high performance expected from eMBB.

Massive MIMO

Massive Multiple Input Multiple Output (Massive MIMO) employs dozens or even hundreds of antennas at the base station.

Benefits include:

- Higher throughput
- Better spectral efficiency
- Increased network capacity
- Improved signal quality

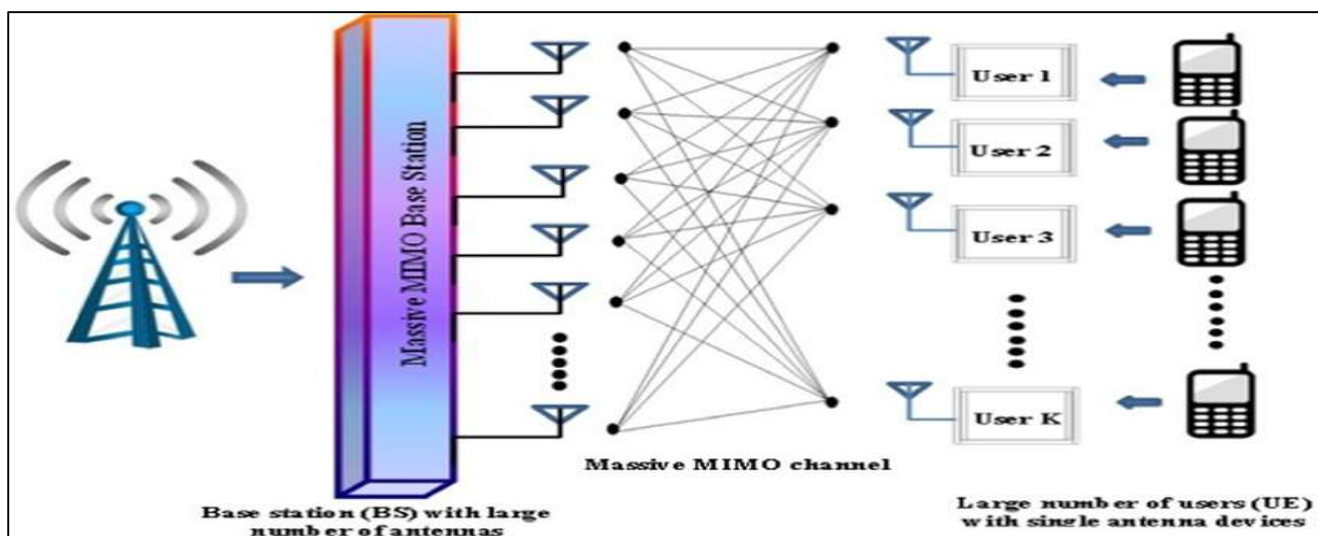


Figure 2.1: Massive MIMO Antenna System

Beamforming

Beamforming directs radio signals toward specific users instead of broadcasting energy uniformly in all directions.

Advantages include:

- Improved signal strength
- Reduced interference
- Better coverage
- Increased network efficiency

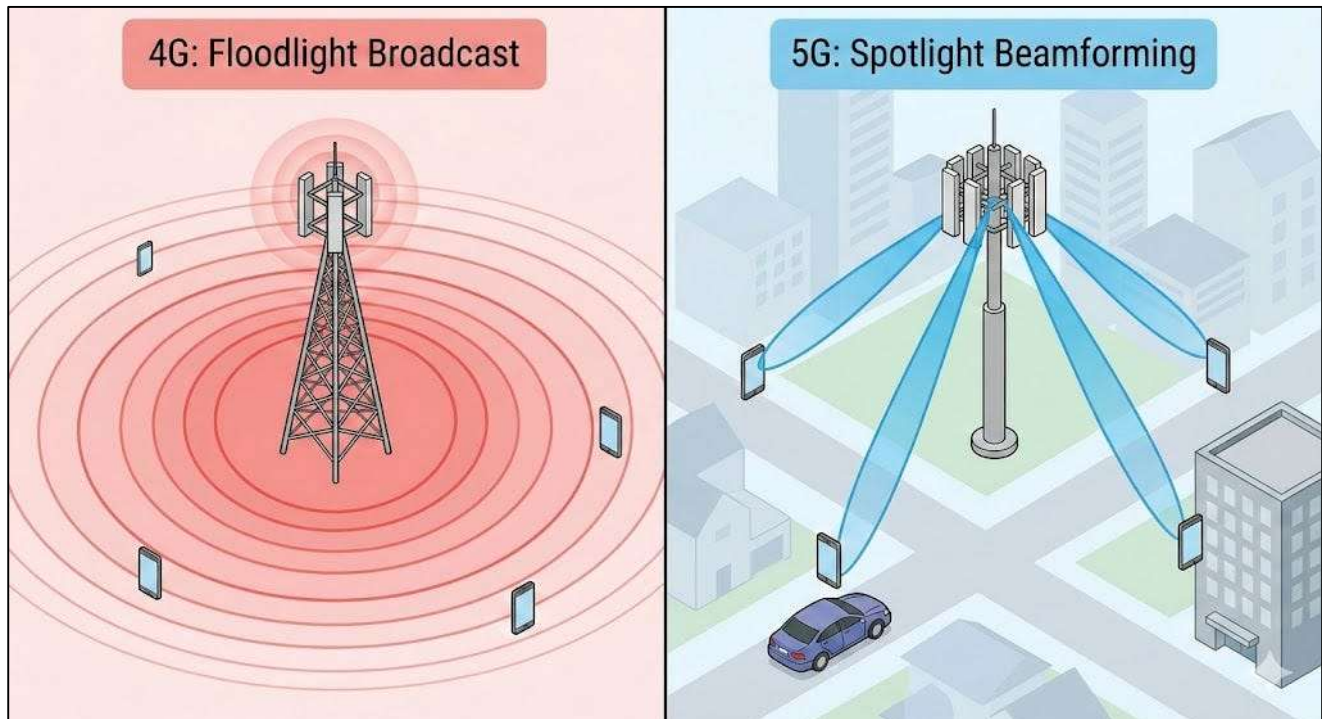


Figure 2.2: Beamforming Concept

Millimeter Wave (mmWave)

Millimeter-wave communication operates at frequencies above 24 GHz and provides very large bandwidths.

Advantages:

- Extremely high data rates
- Large spectrum availability
- Supports multi-gigabit communication

Challenges:

- Short communication range

- High path loss
- Signal blockage by obstacles
- Requirement for dense small-cell deployment

Small Cells

Small cells are low-power base stations deployed in dense urban environments.

Benefits include:

- Increased network capacity
- Better indoor coverage
- Reduced network congestion
- Improved user experience

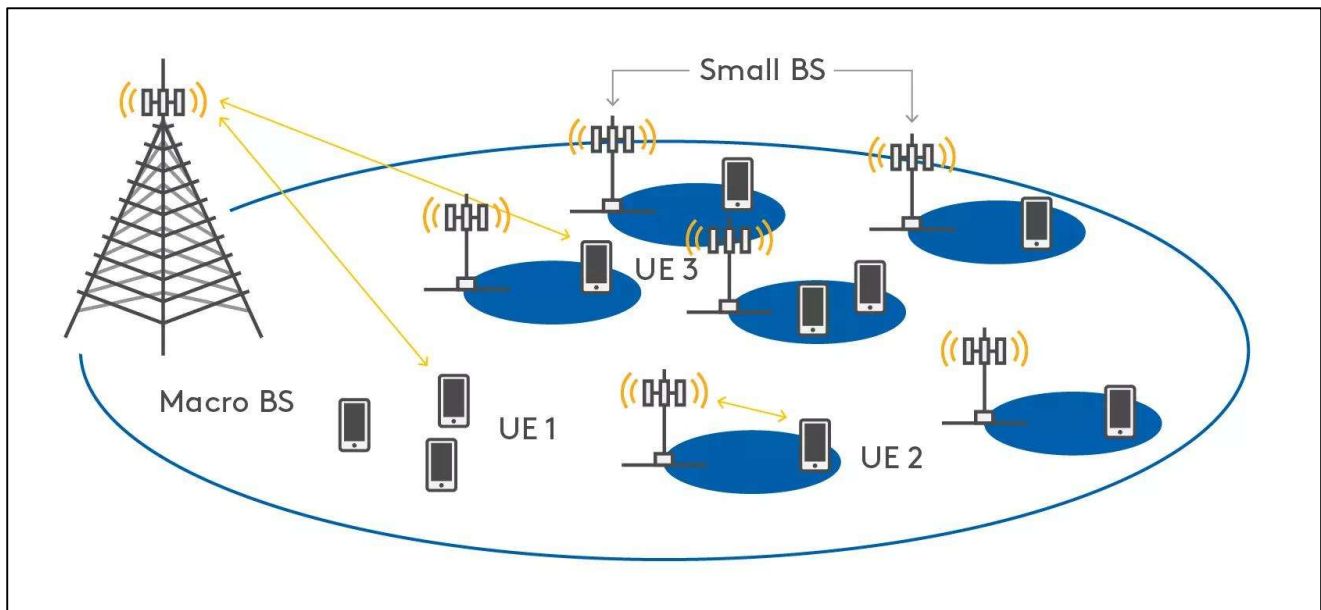


Figure 2.3: Small Cell Deployment

2.1.5 Performance Targets

Typical performance targets for eMBB include:

Parameter	Typical Target
Peak Download Speed	20 Gbps
Peak Upload Speed	10 Gbps
User Experienced Speed	100 Mbps+

Spectral Efficiency	15 3×–5× improvement over LTE
Mobility Support	Up to 500 km/h

2.1.6 Applications of eMBB

eMBB enables a wide range of bandwidth-intensive applications.

Ultra HD Video Streaming

Users can stream 4K and 8K videos with minimal buffering and superior visual quality.

Cloud Gaming

Cloud-based gaming platforms require high bandwidth and low latency to ensure smooth gameplay without downloading large game files.

Examples include:

- Cloud rendering
 - Multiplayer gaming
 - Interactive entertainment
-

Virtual Reality (VR)

Virtual Reality applications require high-resolution displays updated at extremely high frame rates.

Applications include:

- Medical training
 - Education
 - Industrial simulation
 - Gaming
-

Augmented Reality (AR)

AR overlays digital information onto the physical world.

Applications include:

- Navigation

- Manufacturing
 - Maintenance
 - Retail
 - Education
-

Smart Education

Educational institutions benefit from:

- Live interactive lectures
 - Virtual laboratories
 - Remote classrooms
 - High-quality video conferencing
-

Healthcare

Healthcare applications include:

- High-definition telemedicine
 - Medical imaging transfer
 - Remote diagnostics
 - Virtual consultations
-

Smart Manufacturing

Factories utilize eMBB for:

- Digital twins
 - Industrial monitoring
 - High-definition surveillance
 - Remote equipment control
-

2.1.7 Advantages of eMBB

Major advantages include:

- Extremely high data rates.

- Better user experience.
 - High network capacity.
 - Reduced congestion.
 - Improved multimedia quality.
 - Efficient spectrum utilization.
 - Reliable connectivity in crowded environments.
-

2.1.8 Limitations

Despite its advantages, eMBB also has certain limitations:

- High deployment cost.
 - Dependence on dense infrastructure.
 - Limited mmWave propagation.
 - Indoor penetration challenges.
 - Higher power consumption for some deployments.
-

2.1.9 Relevance to Intelligent Transportation Systems

Although eMBB primarily focuses on high-bandwidth communication, it also contributes to intelligent transportation by enabling services such as:

- High-definition traffic camera streaming.
- Real-time road monitoring.
- Connected infotainment systems.
- Vehicle software updates.
- High-resolution map downloads.
- Passenger multimedia services.

However, mission-critical traffic management applications such as emergency vehicle preemption require not only high bandwidth but also extremely low latency and ultra-high reliability. These requirements are addressed by **Ultra-Reliable Low-Latency Communication (URLLC)**, which is discussed in the next section.

2.1.10 Sectional Summary

Enhanced Mobile Broadband forms the foundation for high-speed wireless communication in 5G networks. By combining technologies such as Massive MIMO, Beamforming, Small Cells, and mmWave communication, eMBB delivers multi-gigabit data rates, improved network capacity, and enhanced user experiences. It enables numerous applications including cloud gaming, virtual reality, telemedicine, industrial automation, and smart education. While eMBB addresses bandwidth-intensive applications, latency-sensitive services such as autonomous vehicles and emergency vehicle traffic signal preemption require the capabilities of URLLC, which will be discussed in the following section.

2.2 Ultra-Reliable Low-Latency Communication (URLLC)

2.2.1 Introduction

Ultra-Reliable Low-Latency Communication (URLLC) is one of the three fundamental service categories defined for fifth-generation (5G) mobile networks. While Enhanced Mobile Broadband (eMBB) focuses on delivering high data rates and Massive Machine-Type Communication (mMTC) emphasizes large-scale device connectivity, URLLC is specifically designed for applications where **communication reliability and response time are critical**.

In many real-world scenarios, communication delays of even a few milliseconds can result in severe consequences. Applications such as autonomous driving, industrial automation, remote robotic surgery, smart power grids, and emergency response systems require communication networks capable of transmitting information almost instantaneously while maintaining extremely high reliability.

Traditional communication technologies, including 4G LTE, generally provide communication latency between **30 and 50 milliseconds**. Although this performance is sufficient for video streaming, web browsing, and voice communication, it is inadequate for mission-critical systems that require immediate decision-making.

To address these requirements, 5G introduces URLLC, which is capable of delivering end-to-end communication latency as low as **1–5 milliseconds** while maintaining **99.999% communication reliability**. This capability enables intelligent machines, vehicles, sensors, and infrastructure to exchange information almost in real time.

For this internship project, URLLC forms the technological foundation of the proposed emergency vehicle traffic signal preemption system, where reducing communication delay directly contributes to faster emergency response and improved public safety.

2.2.2 Objectives of URLLC

The primary objectives of URLLC are:

- Achieve ultra-low communication latency.
 - Ensure extremely high communication reliability.
 - Support mission-critical applications.
 - Minimize packet loss during communication.
 - Provide deterministic network performance.
 - Enable real-time machine-to-machine communication.
 - Improve operational safety in intelligent systems.
-

2.2.3 Key Characteristics

Ultra-Low Latency

Latency refers to the time required for data to travel from the sender to the receiver.

One of the most significant improvements introduced by URLLC is the reduction of communication latency from approximately **30–50 ms in 4G LTE** to **1–5 ms in 5G**.

Lower latency enables systems to respond almost instantly to changing environmental conditions.

Examples include:

- Emergency vehicle approaching an intersection.
 - Robot detecting an obstacle.
 - Remote surgeon controlling surgical equipment.
 - Autonomous vehicle avoiding a collision.
-

Ultra-High Reliability

Reliability represents the probability that transmitted information successfully reaches its destination without errors.

URLLC targets communication reliability of **99.999%**, often referred to as "**five nines**" **reliability**.

Such reliability is essential because failures in mission-critical systems can result in:

- Road accidents
- Equipment damage
- Industrial failures

- Delayed emergency response
 - Risks to human life
-

High Availability

URLLC networks are designed to remain operational even under heavy traffic loads or challenging communication environments.

High availability ensures uninterrupted communication for:

- Hospitals
 - Emergency services
 - Industrial facilities
 - Smart transportation infrastructure
-

Deterministic Communication

Unlike conventional networks where communication delays may vary depending on congestion, URLLC aims to provide predictable communication performance.

Predictable latency is particularly important for:

- Factory automation
 - Traffic signal control
 - Autonomous driving
 - Power system protection
-

2.2.4 Technologies Enabling URLLC

Several advanced technologies contribute to the performance of URLLC.

Edge Computing

Edge computing processes data closer to the user rather than sending all information to distant cloud servers.

Benefits include:

- Reduced communication delay
- Faster decision-making

- Lower network congestion
- Improved reliability

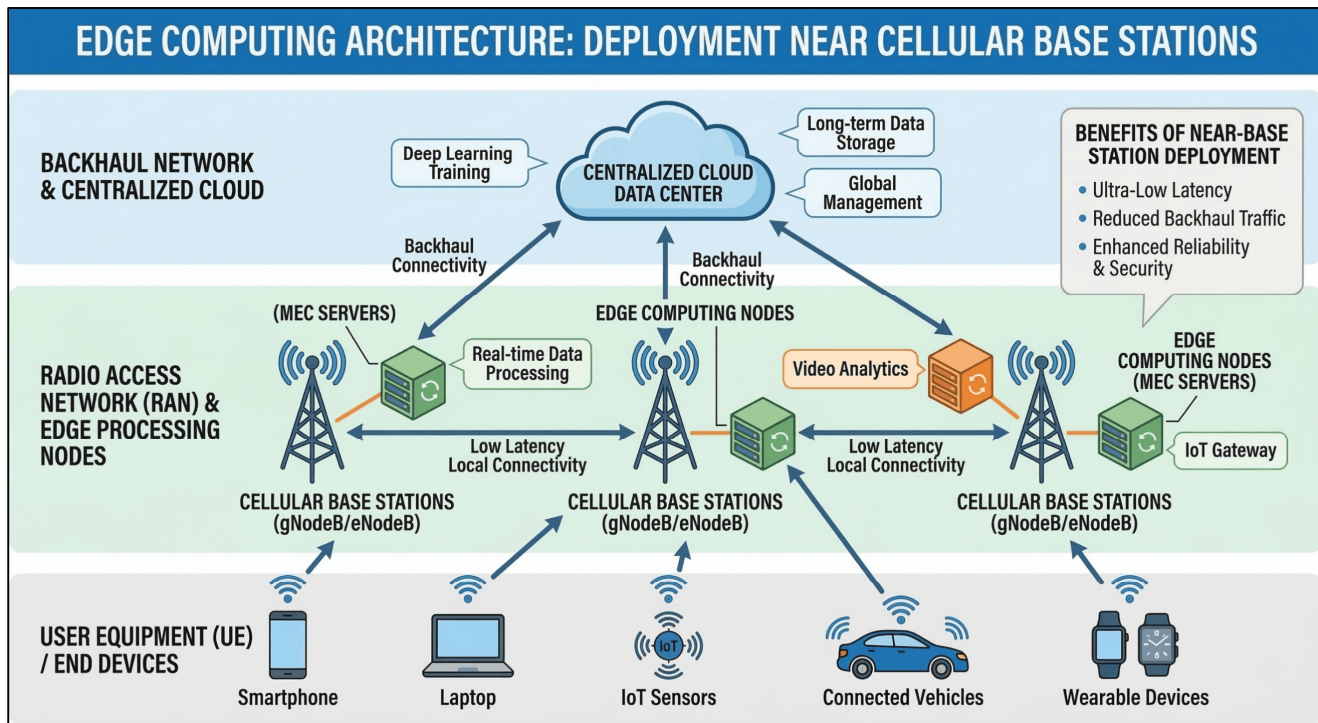


Figure 2.4: Edge Computing Architecture

Network Slicing

Network slicing allows a single physical network to be divided into multiple virtual networks. Each network slice can be optimized for specific applications.

Examples:

- eMBB Slice
- URLLC Slice
- mMTC Slice

This enables emergency communication traffic to receive higher priority than normal Internet traffic.

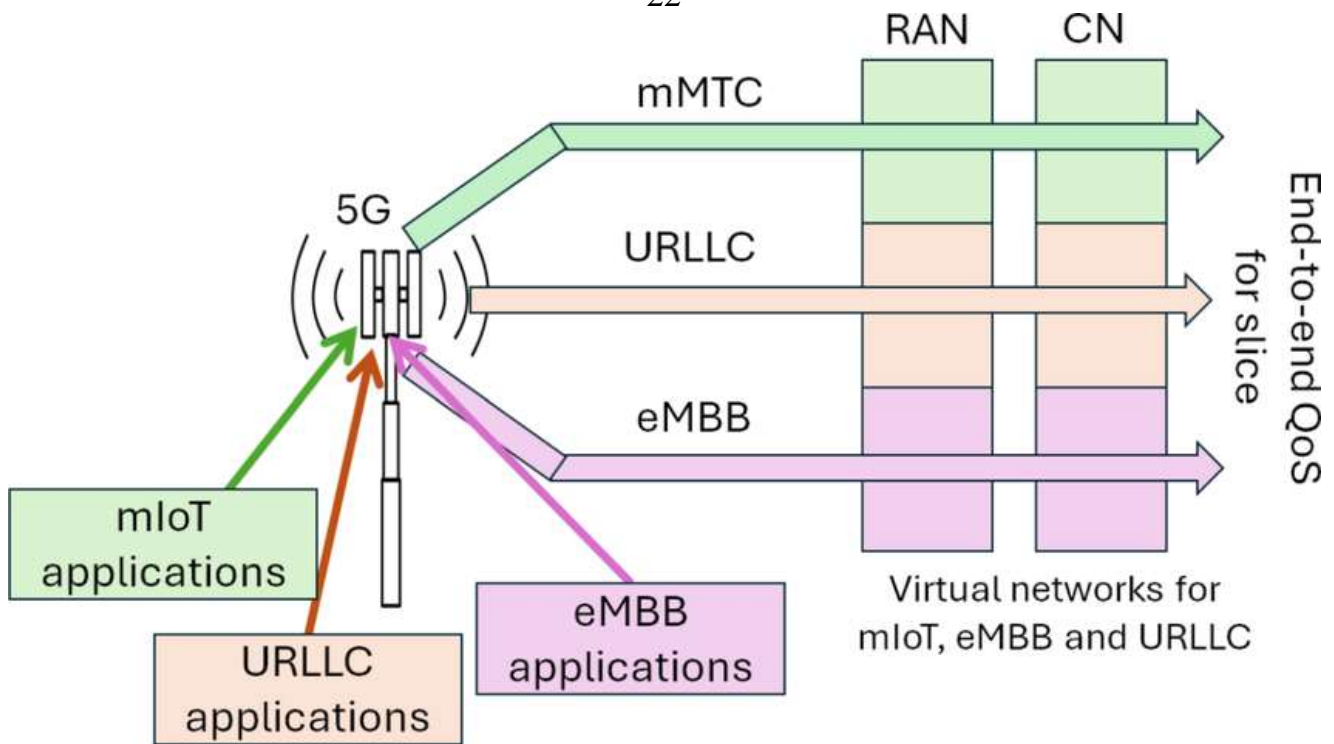


Figure 2.5: Network Slicing Concept

Software Defined Networking (SDN)

SDN separates the network control plane from the data forwarding plane.

Advantages include:

- Centralized management
- Dynamic routing
- Improved resource utilization
- Faster network configuration

Network Function Virtualization (NFV)

NFV replaces dedicated hardware devices with software-based virtual network functions.

Benefits include:

- Reduced infrastructure cost
- Flexible deployment
- Faster service provisioning
- Improved scalability

Artificial Intelligence in Network Management

Artificial Intelligence assists URLLC by:

- Predicting network congestion
 - Optimizing routing decisions
 - Detecting communication failures
 - Allocating network resources efficiently
-

2.2.5 Performance Targets

Typical URLLC performance requirements are summarized below.

Parameter	Target Value
End-to-End Latency	1–5 ms
Reliability	99.999%
Packet Error Rate	Very Low
Availability	Extremely High
Communication Type	Mission Critical

2.2.6 Applications of URLLC

Autonomous Vehicles

Autonomous vehicles continuously exchange information with nearby vehicles, traffic signals, pedestrians, and cloud infrastructure.

Examples include:

- Collision avoidance
 - Lane change assistance
 - Adaptive cruise control
 - Cooperative driving
-

Emergency Vehicle Communication

Emergency vehicles require immediate priority at signalized intersections.

Using URLLC:

- Ambulances transmit their location continuously.
- Traffic controllers receive the request instantly.
- Green signals are activated before vehicle arrival.
- Travel delay is minimized.

This application directly corresponds to the internship project.

Remote Healthcare

Medical applications include:

- Robotic surgery
- Remote patient monitoring
- Emergency consultation
- Medical imaging

Any communication delay could affect patient safety.

Industrial Automation

Modern factories require precise synchronization between robots and production equipment.

Applications include:

- Assembly robots
 - Automated guided vehicles
 - Process control
 - Smart manufacturing
-

Smart Power Grid

Electrical power systems require immediate communication for:

- Fault isolation
- Load balancing
- Grid protection

- Renewable energy integration
-

Railway Safety

Railway communication systems use URLLC for:

- Train control
 - Collision prevention
 - Automatic braking
 - Signaling
-

2.2.7 URLLC in Intelligent Transportation Systems (ITS)

Intelligent Transportation Systems integrate communication, sensing, computing, and automation to improve road safety and traffic efficiency.

URLLC enables:

- Real-time traffic monitoring.
- Adaptive traffic signal control.
- Vehicle-to-Infrastructure (V2I) communication.
- Vehicle-to-Vehicle (V2V) communication.
- Cooperative driving.
- Emergency vehicle priority.

These capabilities reduce traffic congestion while improving road safety.

2.2.8 Cellular Vehicle-to-Everything (C-V2X)

Cellular Vehicle-to-Everything (C-V2X) is a communication framework that enables vehicles to exchange information with surrounding entities.

The major communication types include:

Vehicle-to-Vehicle (V2V)

Communication between nearby vehicles.

Applications:

- Collision avoidance

- Speed synchronization
 - Cooperative lane changing
-

Vehicle-to-Infrastructure (V2I)

Communication between vehicles and roadside infrastructure.

Examples:

- Traffic signals
- Smart intersections
- Roadside units

This communication is essential for emergency vehicle preemption.

Vehicle-to-Network (V2N)

Vehicles communicate with cloud servers through cellular networks.

Applications include:

- Navigation
 - Traffic updates
 - Fleet management
-

Vehicle-to-Pedestrian (V2P)

Communication with smartphones and wearable devices improves pedestrian safety.

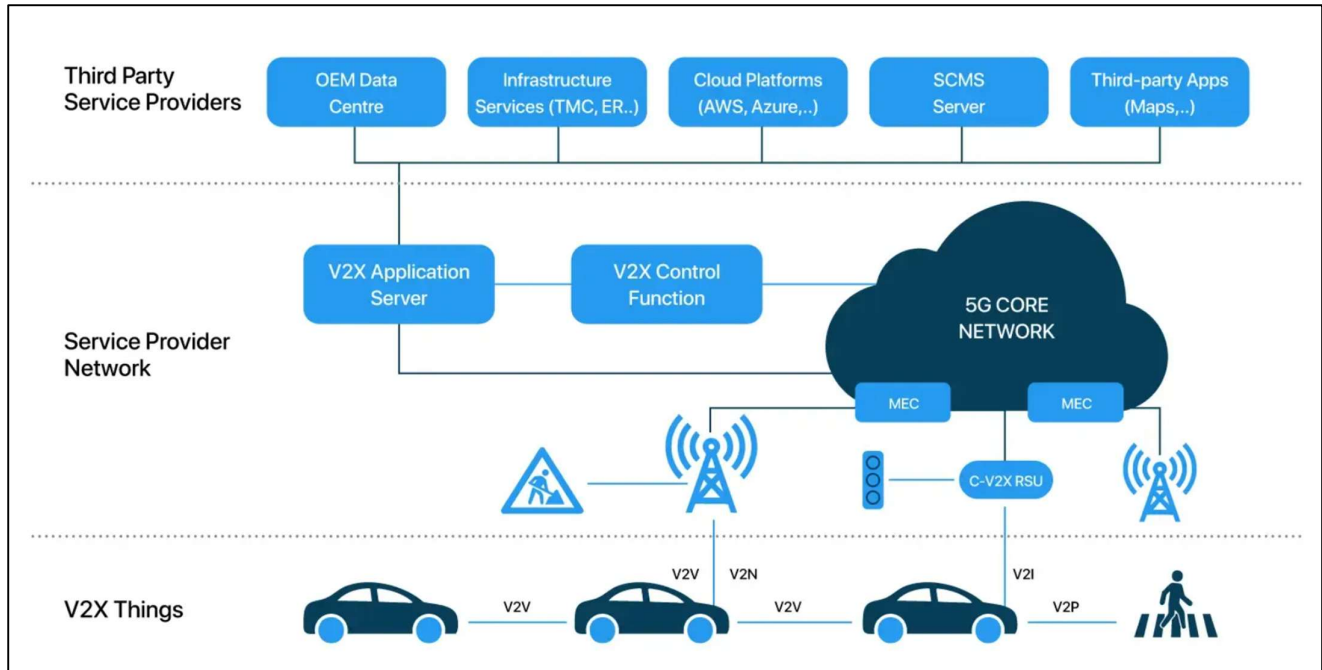


Figure 2.6: C-V2X Communication Architecture

2.2.9 URLLC for Emergency Vehicle Preemption

One of the most important applications of URLLC is emergency vehicle traffic signal preemption.

In conventional systems:

1. Emergency vehicle approaches an intersection.
2. Traffic signal follows fixed timing.
3. Vehicle waits at the red signal.
4. Response time increases.

Using 5G URLLC:

1. Emergency vehicle continuously transmits its location.
2. Traffic controller receives the information within a few milliseconds.
3. Traffic light phase changes automatically.
4. Downstream traffic begins clearing.
5. Emergency vehicle crosses without stopping.
6. Original signal timing resumes after passage.

This significantly reduces:

- Travel time

- Waiting time
- Queue length
- Fuel consumption
- Risk of accidents

2.2.10 Comparison of 4G LTE and 5G URLLC

<i>Parameter</i>	<i>4G LTE</i>	<i>5G URLLC</i>
<i>Latency</i>	30–50 ms	1–5 ms
<i>Reliability</i>	High	99.999%
<i>Response Speed</i>	Moderate	Extremely Fast
<i>Mission-Critical Support</i>	Limited	Excellent
<i>Emergency Vehicle Applications</i>	Possible	Highly Suitable
<i>Industrial Automation</i>	Limited	Fully Supported
<i>Autonomous Driving</i>	Partial	Full Support

2.2.11 Relevance to the Internship Project

The internship titled "**Development of Vehicle-to-Infrastructure (V2I)-Based Traffic Signal Preemption for Emergency Vehicles for Emergency Vehicles**" directly utilizes the principles of URLLC.

In the developed simulation:

- **SUMO** models urban traffic conditions.
- **Python TraCI** dynamically controls traffic signal phases.
- Three scenarios are evaluated:
 - Baseline (No Preemption)
 - 4G LTE Preemption (30–50 ms latency)
 - 5G URLLC Preemption (1–5 ms latency)

The objective is to demonstrate how lower communication latency enables earlier traffic signal activation, resulting in faster emergency vehicle movement and improved traffic management under varying congestion levels.

2.2.12 Sectional Summary

Ultra-Reliable Low-Latency Communication (URLLC) is one of the most transformative capabilities of 5G technology. By providing communication latency as low as 1–5 milliseconds with 99.999% reliability, URLLC enables a wide range of mission-critical applications including autonomous transportation, industrial automation, healthcare, and emergency response systems. Through technologies such as edge computing, network slicing, SDN, NFV, and C-V2X, URLLC establishes the communication foundation required for intelligent transportation systems. In the context of this internship, URLLC enables efficient emergency vehicle traffic signal preemption by ensuring rapid and reliable communication between emergency vehicles and traffic infrastructure, ultimately reducing response times and enhancing public safety.

2.3 Massive Machine-Type Communication (mMTC)

2.3.1 Introduction

Massive Machine-Type Communication (mMTC) is the third major service category defined in the 5G communication framework. While Enhanced Mobile Broadband (eMBB) focuses on providing high-speed data services and Ultra-Reliable Low-Latency Communication (URLLC) supports mission-critical applications, mMTC is specifically designed to enable the large-scale deployment of Internet of Things (IoT) devices.

The rapid growth of smart cities, industrial automation, precision agriculture, environmental monitoring, healthcare, and intelligent transportation systems has led to an unprecedented increase in the number of connected devices. Unlike smartphones, most IoT devices transmit relatively small amounts of data but require long battery life, reliable connectivity, and the ability to coexist with millions of similar devices.

Traditional communication networks were primarily designed for human-to-human communication and are not capable of efficiently managing such an enormous number of connected sensors and machines. To address this challenge, 5G introduces mMTC, enabling networks to support **up to one million connected devices per square kilometer** while maintaining efficient spectrum utilization and low power consumption.

2.3.2 Objectives of mMTC

The primary objectives of Massive Machine-Type Communication include:

- Support an extremely large number of connected devices.
- Provide energy-efficient communication.
- Enable long battery life for IoT sensors.
- Improve network scalability.

- Reduce deployment and maintenance costs.
 - Support wide-area connectivity.
 - Facilitate large-scale automation in smart environments.
-

2.3.3 Characteristics of mMTC

Massive Connectivity

One of the defining features of mMTC is its ability to support a very high density of connected devices.

Typical deployment scenarios include:

- Smart cities
- Industrial automation
- Smart agriculture
- Environmental monitoring
- Utility networks

Unlike conventional mobile networks, which mainly serve human users, mMTC focuses on machine-to-machine (M2M) communication.

Low Power Consumption

Many IoT devices operate using batteries and are deployed in remote or inaccessible locations. Frequent battery replacement is often impractical.

mMTC communication protocols are therefore optimized to minimize energy consumption, enabling devices to operate continuously for several years without battery replacement.

Typical battery life ranges from **5 to 10 years**, depending on the application.

Small Data Transmission

Most IoT devices periodically transmit small packets of information rather than continuous multimedia streams.

Examples include:

- Temperature readings
- Humidity measurements

- Air quality data
- Vehicle location updates
- Energy meter readings

This allows the network to efficiently accommodate a large number of devices simultaneously.

Wide Area Coverage

Many IoT deployments span large geographical regions.

Examples include:

- Agricultural fields
- Smart power grids
- Oil pipelines
- Environmental monitoring stations
- Transportation corridors

mMTC provides reliable connectivity even in challenging environments where conventional broadband coverage may be limited.

Cost-Effective Deployment

IoT devices are generally inexpensive and widely distributed.

mMTC reduces deployment costs through:

- Simplified communication protocols.
 - Energy-efficient operation.
 - Low-complexity device hardware.
 - Scalable network management.
-

2.3.4 Technologies Supporting mMTC

Several technologies contribute to the implementation of mMTC within 5G networks.

Narrowband IoT (NB-IoT)

NB-IoT is designed specifically for low-power wide-area communication.

Features include:

- Excellent indoor coverage.
- Low energy consumption.
- Low deployment cost.
- Long battery life.

Applications include:

- Smart electricity meters
 - Water meters
 - Gas meters
 - Environmental sensors
-

LTE-M

LTE-M provides improved mobility compared to NB-IoT while maintaining low power consumption.

Applications include:

- Wearable devices
 - Asset tracking
 - Connected vehicles
 - Smart healthcare
-

Low Power Wide Area Networks (LPWAN)

LPWAN technologies enable long-distance communication while consuming minimal power.

Characteristics include:

- Long communication range.
 - Low operating cost.
 - High scalability.
 - Low energy consumption.
-

2.3.5 Performance Targets

Typical performance objectives of mMTC are summarized below.

Parameter	Target Value
Device Density	Up to 1 million devices/km ²
Battery Life	5–10 years
Data Rate	Low
Power Consumption	Very Low
Communication Type	Machine-to-Machine

2.3.6 Applications of mMTC

Smart Cities

mMTC enables intelligent city infrastructure through interconnected sensors and monitoring systems.

Applications include:

- Smart street lighting
 - Smart parking
 - Waste management
 - Environmental monitoring
 - Intelligent traffic monitoring
-

Smart Agriculture

Agricultural IoT systems use mMTC for:

- Soil moisture monitoring
 - Weather monitoring
 - Smart irrigation
 - Livestock tracking
 - Crop health analysis
-

Industrial Automation

Factories deploy thousands of sensors to monitor equipment and production processes.

Applications include:

- Machine condition monitoring
 - Predictive maintenance
 - Inventory management
 - Automated production systems
-

Healthcare

Healthcare IoT devices continuously monitor patients.

Examples include:

- Heart rate sensors
 - Blood pressure monitors
 - Glucose monitoring
 - Wearable health devices
-

Smart Energy

Power utilities utilize mMTC for:

- Smart electricity meters
 - Grid monitoring
 - Renewable energy integration
 - Fault detection
-

Environmental Monitoring

Government agencies deploy IoT sensors to monitor:

- Air pollution
 - Water quality
 - Temperature
 - Noise levels
 - Flood warning systems
-

2.3.7 mMTC in Intelligent Transportation Systems

Intelligent transportation systems require thousands of interconnected sensors distributed throughout road infrastructure.

Examples include:

- Traffic volume sensors.
- Vehicle counting systems.
- Parking sensors.
- Road weather stations.
- Air quality monitoring.
- Smart street lights.

These devices continuously collect data that enables authorities to optimize traffic flow and improve transportation efficiency.

2.3.8 Relationship Between mMTC and Emergency Vehicle Preemption

Although the proposed internship primarily utilizes URLLC for emergency vehicle communication, mMTC provides valuable supporting infrastructure.

Examples include:

- Traffic density sensors.
- Roadside environmental sensors.
- Smart parking systems.
- Road occupancy detectors.
- Connected surveillance cameras.
- Smart pedestrian crossings.

Information collected by these devices can assist traffic management systems in making more informed decisions during emergency vehicle routing.

For example, traffic density sensors may identify heavily congested intersections, allowing traffic controllers to adjust signal timing before an emergency vehicle reaches the affected area.

2.3.9 Advantages of mMTC

Major advantages include:

- Supports millions of connected devices.
- Low power consumption.
- Long battery life.
- Low deployment cost.
- High scalability.
- Efficient spectrum utilization.
- Enables smart city infrastructure.
- Supports large-scale IoT ecosystems.

2.3.10 Limitations of mMTC

Despite its advantages, mMTC also has several limitations:

- Low data transmission rates.
- Not suitable for multimedia applications.
- Higher communication latency compared to URLLC.
- Limited support for mission-critical applications.
- Potential security and privacy concerns in large IoT deployments.

2.3.11 Comparison of eMBB, URLLC and mMTC

The three primary service categories of 5G are designed to satisfy different communication requirements.

<i>Feature</i>	<i>eMBB</i>	<i>URLLC</i>	<i>mMTC</i>
<i>Primary Objective</i>	High-Speed Broadband	Ultra-Reliable Low-Latency Communication	Massive IoT Connectivity
<i>Typical Latency</i>	Moderate	1–5 ms	Moderate
<i>Reliability</i>	High	Very High (99.999%)	High
<i>Data Rate</i>	Very High	Moderate	Low

<i>Device Density</i>	Moderate	Moderate	Extremely High
<i>Main Applications</i>	Video Streaming, VR, Gaming	Autonomous Vehicles, Healthcare, Emergency Systems	Smart Cities, Agriculture, IoT Sensors

2.3.12 Relevance to the Internship Project

The internship project focuses on **Development of Vehicle-to-Infrastructure (V2I)-Based Traffic Signal Preemption for Emergency Vehicles**. While URLLC provides the low-latency communication required for dynamic traffic signal control, mMTC complements the system by enabling large-scale deployment of roadside sensing infrastructure.

In future smart transportation systems, information collected from thousands of IoT devices—including traffic detectors, environmental sensors, and smart infrastructure—can be integrated with URLLC communication to create more intelligent and adaptive traffic management solutions.

Such integration will enable emergency vehicle routing decisions based not only on vehicle location but also on real-time traffic density, road conditions, weather information, and infrastructure status.

2.3.13 Sectional Summary

Massive Machine-Type Communication (mMTC) enables 5G networks to support millions of interconnected IoT devices with low power consumption, long battery life, and scalable communication. By facilitating large-scale deployment of sensors and smart infrastructure, mMTC plays a vital role in applications such as smart cities, healthcare, industrial automation, agriculture, environmental monitoring, and intelligent transportation systems. Although emergency vehicle preemption primarily depends on URLLC for ultra-fast communication, mMTC provides the supporting sensor ecosystem necessary for future intelligent traffic management. Together, eMBB, URLLC, and mMTC form the foundation of the versatile and application-oriented architecture of fifth-generation mobile communication networks.

2.4 Overall Chapter Summary

This chapter presented the three fundamental service categories of 5G: **Enhanced Mobile Broadband (eMBB)**, **Ultra-Reliable Low-Latency Communication (URLLC)**, and **Massive Machine-Type Communication (mMTC)**. Each category addresses distinct communication requirements, ranging from high-speed broadband services to mission-critical real-time communication and large-scale IoT connectivity. The chapter also highlighted the

importance of URLLC in enabling emergency vehicle traffic signal preemption, which serves as the foundation of the internship project. The next chapter discusses the overall **5G Network Architecture**, including the 5G Core (5GC), Radio Access Network (RAN), gNodeB (gNB), network slicing, Massive MIMO, beamforming, and spectrum allocation.

Chapter 3

5G Network Architecture

The fifth generation (5G) mobile communication system introduces a highly flexible, software-driven, and service-oriented network architecture designed to support diverse applications ranging from enhanced mobile broadband (eMBB) to ultra-reliable low-latency communication (URLLC) and massive machine-type communication (mMTC). Unlike previous cellular generations, which primarily focused on increasing data rates, the 5G architecture has been developed to provide high scalability, network intelligence, virtualization, and dynamic resource allocation.

The overall 5G architecture consists of two primary components:

- **5G Radio Access Network (5G RAN)**
- **5G Core Network (5GC)**

These components work together to provide seamless communication between user devices, network infrastructure, cloud services, and application servers.

3.1 Overview of 5G Architecture

The architecture of a typical 5G network can be represented as follows:

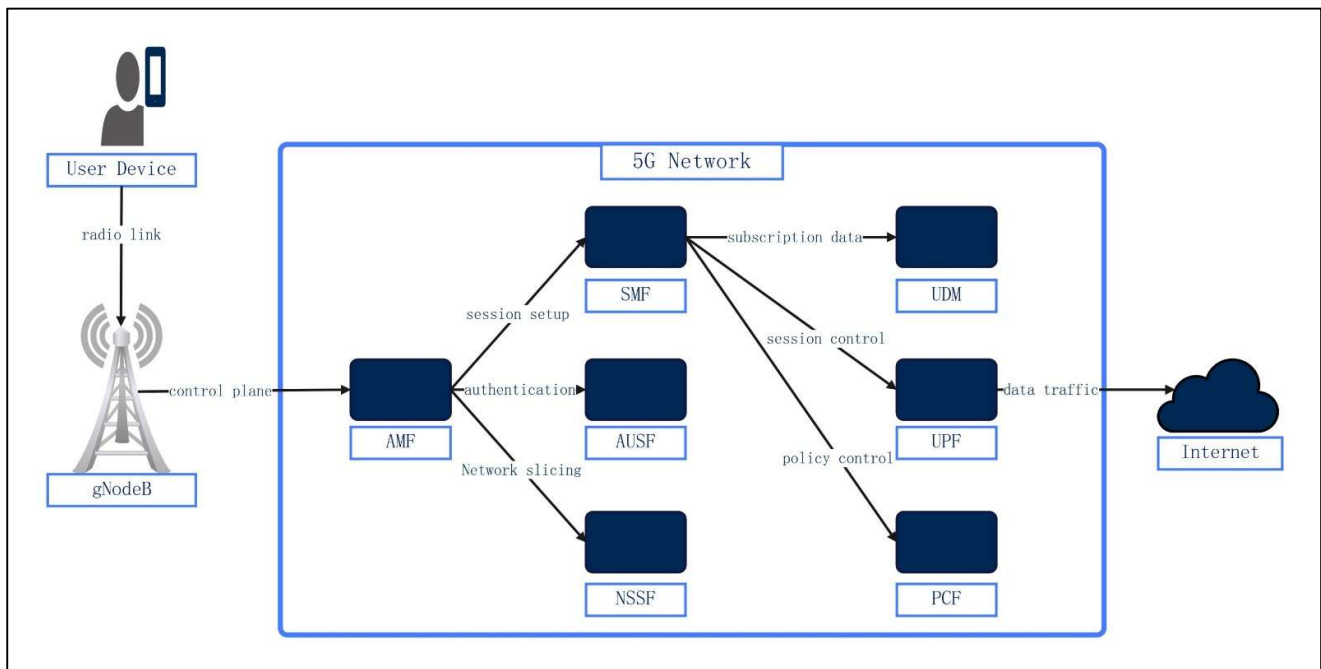


Figure 3.1: Overall 5G Network Architecture

3.2 User Equipment (UE)

User Equipment (UE) refers to all devices that access the 5G network. These devices communicate wirelessly with the nearest gNodeB (gNB), which acts as the radio access point.

Examples of User Equipment include:

- Smartphones
- Tablets
- Laptops
- Smartwatches
- IoT devices
- Autonomous vehicles
- Emergency vehicles
- Industrial robots
- Medical devices

In the context of this internship, the emergency vehicle can be considered as a connected User Equipment that continuously exchanges information with roadside infrastructure through the 5G network.

3.3 5G Radio Access Network (RAN)

The Radio Access Network (RAN) provides the wireless communication interface between user devices and the 5G Core Network.

Its primary responsibilities include:

- Radio signal transmission
- User scheduling
- Resource allocation
- Mobility management
- Signal quality monitoring
- Handover management

Compared with LTE, the 5G RAN supports significantly higher bandwidth, improved spectral efficiency, lower latency, and better network capacity.

3.4 gNodeB (gNB)

The gNodeB, commonly abbreviated as **gNB**, is the 5G equivalent of the LTE eNodeB.

It performs several important functions including:

- Wireless communication with user devices.
- Radio resource management.
- Scheduling of data transmission.
- Beamforming control.
- Massive MIMO processing.
- Forwarding traffic to the 5G Core.

The gNB is also responsible for establishing and maintaining communication links with moving users, including connected vehicles.

3.5 5G Core Network (5GC)

The 5G Core Network represents the intelligence of the entire communication system.

Unlike previous generations, the 5G Core adopts a **Service-Based Architecture (SBA)** where different network functions communicate using standardized interfaces.

Major functions of the 5G Core include:

- Authentication
- Session management
- User mobility
- Data routing
- Policy control
- Network slicing
- Quality of Service (QoS)
- Security management

The modular architecture enables operators to deploy, upgrade, and scale network functions independently.

3.6 Major Functional Components of the 5G Core

3.6.1 Access and Mobility Management Function (AMF)

The AMF manages:

- User registration
- Authentication
- Mobility management
- Device connectivity
- Handover procedures

Whenever a mobile user moves between different cells, the AMF ensures uninterrupted communication.

3.6.2 Session Management Function (SMF)

The SMF establishes and manages communication sessions.

Its responsibilities include:

- IP address allocation
 - Session creation
 - Session modification
 - Session termination
 - Traffic routing policies
-

3.6.3 User Plane Function (UPF)

The UPF is responsible for forwarding user data packets between the radio network and external data networks.

Functions include:

- Packet forwarding
- Packet inspection
- Quality of Service enforcement
- Traffic routing

Since user data flows through the UPF, placing it closer to users through edge computing significantly reduces communication latency.

3.6.4 Policy Control Function (PCF)

The PCF determines network policies such as:

- Quality of Service
- Charging rules
- Traffic prioritization
- Resource allocation

Mission-critical traffic such as emergency vehicle communication may receive higher priority through policy control.

3.6.5 Authentication Server Function (AUSF)

The AUSF authenticates users before network access.

Its objectives include:

- Prevent unauthorized access.
 - Improve network security.
 - Protect user identity.
-

3.6.6 Unified Data Management (UDM)

The UDM stores subscriber-related information including:

- User profile
 - Authentication credentials
 - Subscription details
 - Service permissions
-

3.7 Service-Based Architecture (SBA)

One of the major innovations in the 5G Core is the Service-Based Architecture.

Instead of tightly coupled hardware components, network functions communicate using standardized software interfaces.

Advantages include:

- Flexible deployment
- Easier maintenance
- Better scalability
- Cloud-native implementation
- Faster software updates

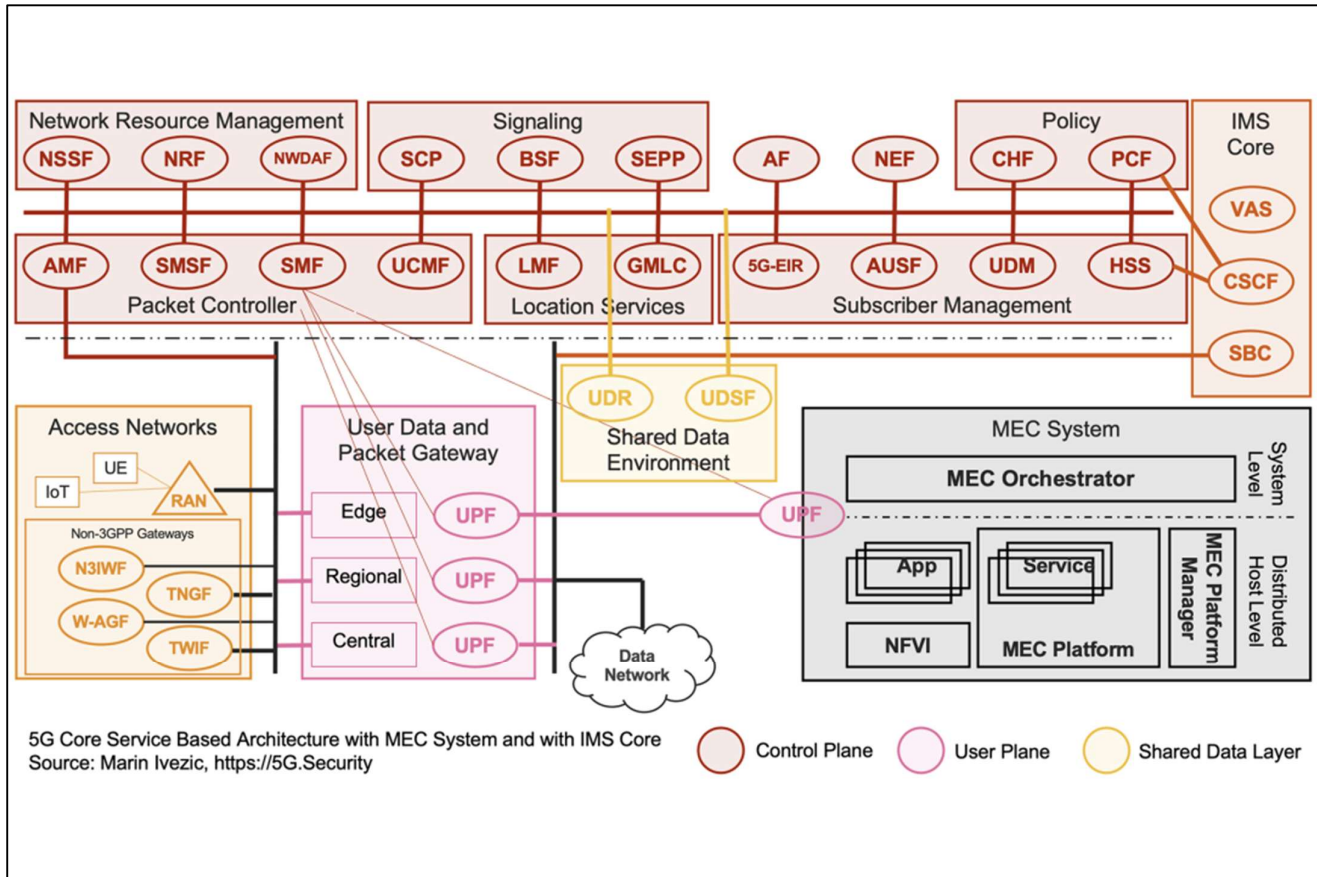


Figure 3.2: Service-Based Architecture (SBA) of the 5G Core

3.8 Network Slicing

Network slicing enables multiple logical networks to operate on the same physical infrastructure.

Each slice is customized according to application requirements.

Examples include:

Slice Type	Application
eMBB Slice	High-speed Internet
URLLC Slice	Emergency services

For emergency vehicle preemption, a dedicated URLLC slice ensures that communication is not delayed by ordinary Internet traffic.

3.9 Multi-Access Edge Computing (MEC)

Edge computing places computational resources close to users rather than relying solely on distant cloud servers.

Benefits include:

- Lower latency
- Reduced backhaul traffic
- Faster response
- Improved reliability

For intelligent transportation systems, edge servers may be installed near major intersections to process emergency vehicle requests almost instantly.

3.10 Network Function Virtualization (NFV)

Traditional communication networks rely heavily on specialized hardware.

NFV replaces these dedicated hardware appliances with software-based virtual machines.

Advantages include:

- Lower deployment cost
 - Flexible upgrades
 - Improved scalability
 - Efficient resource utilization
-

3.11 Software Defined Networking (SDN)

Software Defined Networking separates the network control functions from the forwarding functions.

Benefits include:

- Centralized network management

- Dynamic routing
 - Intelligent traffic engineering
 - Faster service deployment
-

3.12 Security in 5G Networks

Security is a fundamental requirement of 5G communication systems.

Important security mechanisms include:

- Subscriber authentication
- End-to-end encryption
- User identity protection
- Secure communication channels
- Network slicing isolation
- Intrusion detection systems

These mechanisms are particularly important for mission-critical applications such as healthcare, industrial automation, and emergency transportation.

3.13 Integration with Intelligent Transportation Systems

One of the most important applications of the 5G architecture is Intelligent Transportation Systems (ITS).

In such systems:

1. Emergency Vehicle transmits its GPS position.
2. Data is received by the nearest gNodeB.
3. Information reaches the 5G Core.
4. Traffic controller processes the request.
5. Traffic signal changes to green.
6. Emergency vehicle passes safely.
7. Normal signal timing resumes.

This complete communication cycle occurs within a few milliseconds using URLLC.

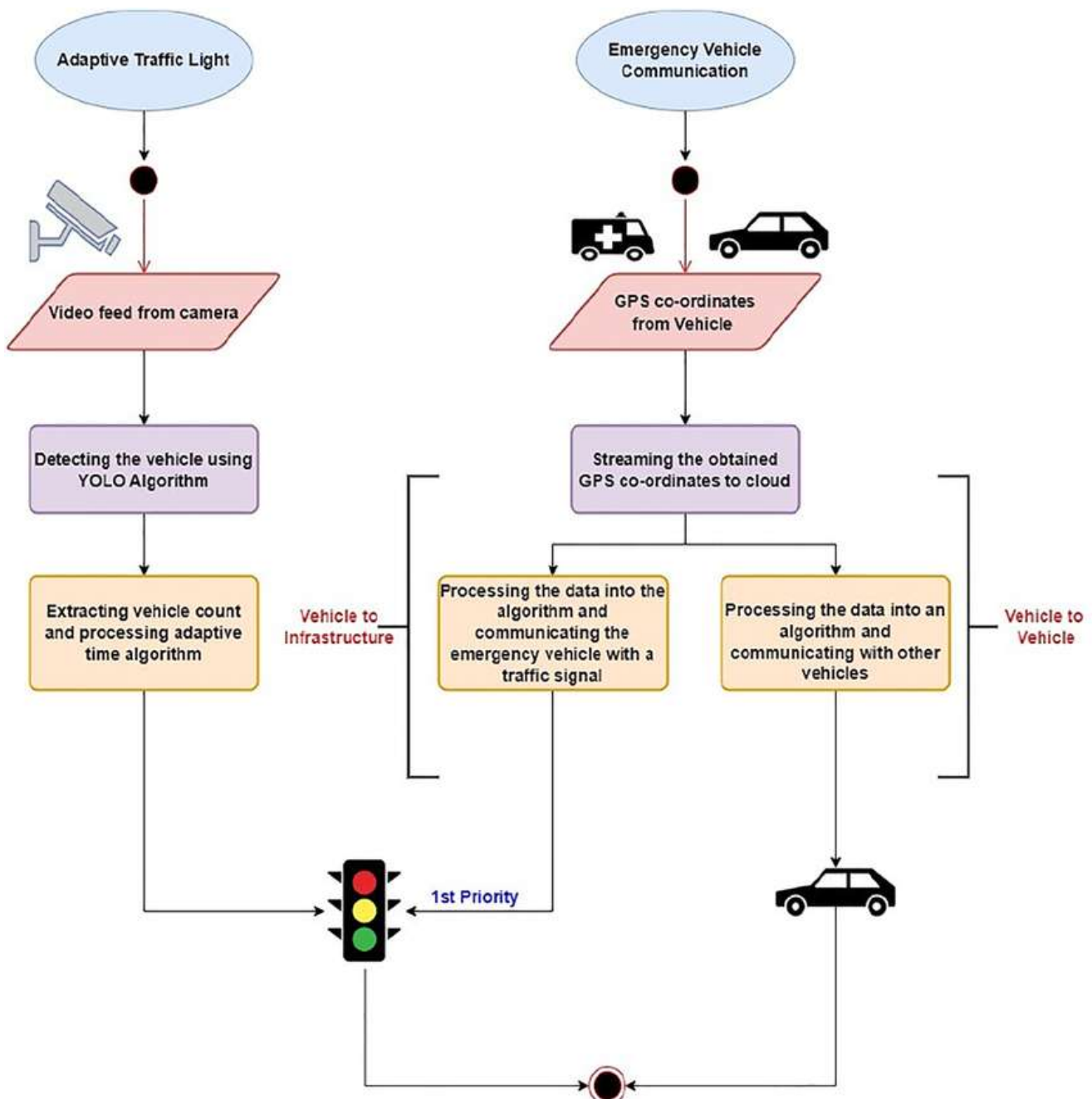


Figure 3.3: Communication Flow for Emergency Vehicle Preemption

3.14 Relevance to the Internship Project

The internship project titled "Development of Vehicle-to-Infrastructure (V2I)-Based Traffic Signal Preemption for Emergency Vehicles" closely follows the communication architecture discussed in this chapter.

Within the simulation:

- **SUMO** represents the road network and traffic environment.
- **Python TraCI** acts as the traffic signal controller.

- The emergency vehicle periodically communicates its position.
- The simulated communication delay represents either 4G LTE or 5G URLLC.
- The traffic signal controller grants signal priority based on the received information.

Although the project is simulation-based, the communication workflow accurately reflects the operation of real-world intelligent transportation systems deployed over 5G infrastructure.

3.15 Chapter Summary

This chapter introduced the architecture of the fifth-generation mobile communication network. The major components of the 5G Radio Access Network, gNodeB, and the 5G Core Network were discussed along with important functional entities such as the AMF, SMF, UPF, PCF, AUSF, and UDM. The concepts of Service-Based Architecture, Network Slicing, Edge Computing, NFV, and SDN were also presented to illustrate how modern 5G networks achieve flexibility, scalability, and ultra-low latency. Finally, the chapter demonstrated how these architectural components support intelligent transportation systems and provide the communication foundation for the proposed emergency vehicle preemption system implemented using SUMO and TraCI. The next chapter discusses the standards and specifications governing the deployment and operation of 5G networks.

Chapter 4

5G Standards

The rapid development and global deployment of fifth-generation (5G) mobile communication networks require internationally accepted standards to ensure interoperability, compatibility, security, and consistent performance across different countries and manufacturers. Standardization enables network operators, equipment vendors, and device manufacturers to develop products that work seamlessly within a unified communication ecosystem.

Several international organizations contribute to the development of 5G standards. Among these, the **3rd Generation Partnership Project (3GPP)** is responsible for defining the technical specifications of 5G New Radio (NR) and the 5G Core Network, while the **International Telecommunication Union (ITU)** establishes the overall performance requirements under the **IMT-2020 (International Mobile Telecommunications-2020)** framework.

These standards ensure that 5G networks satisfy the requirements of high-speed broadband communication, ultra-reliable low-latency communication, and massive machine-type communication while supporting future technological advancements.

4.1 Role of Standardization in Mobile Communication

Standardization is essential for the successful deployment of global communication networks.

The primary objectives of standardization include:

- Ensuring interoperability among equipment from different vendors.
- Providing consistent Quality of Service (QoS).
- Improving communication security.
- Supporting global roaming.
- Enabling efficient spectrum utilization.
- Simplifying network deployment.
- Promoting technological innovation.

Without standardized communication protocols, devices manufactured by different companies would not be able to communicate reliably across various networks.

4.2 Third Generation Partnership Project (3GPP)

4.2.1 Introduction

The **Third Generation Partnership Project (3GPP)** is an international collaboration of telecommunications standard development organizations responsible for developing specifications for GSM, UMTS, LTE, and 5G technologies.

3GPP develops detailed technical specifications covering:

- Radio access technologies.
- Core network architecture.
- Security mechanisms.
- Network protocols.
- Mobility management.
- Quality of Service.
- Spectrum utilization.

Its specifications are widely adopted by network operators and equipment manufacturers worldwide.

4.2.2 Evolution of 3GPP Releases

The development of 5G technology has progressed through multiple standardized releases.

Release 15

Release 15, finalized in 2018, introduced the first commercial version of 5G New Radio (NR).

Major features include:

- 5G NR
- Non-Standalone (NSA) deployment
- Standalone (SA) architecture
- Enhanced Mobile Broadband (eMBB)
- Initial 5G Core architecture

3GPP STANDARD RELEASES TIMELINE: HIGHLIGHTS OF REL-15, REL-16, & REL-17

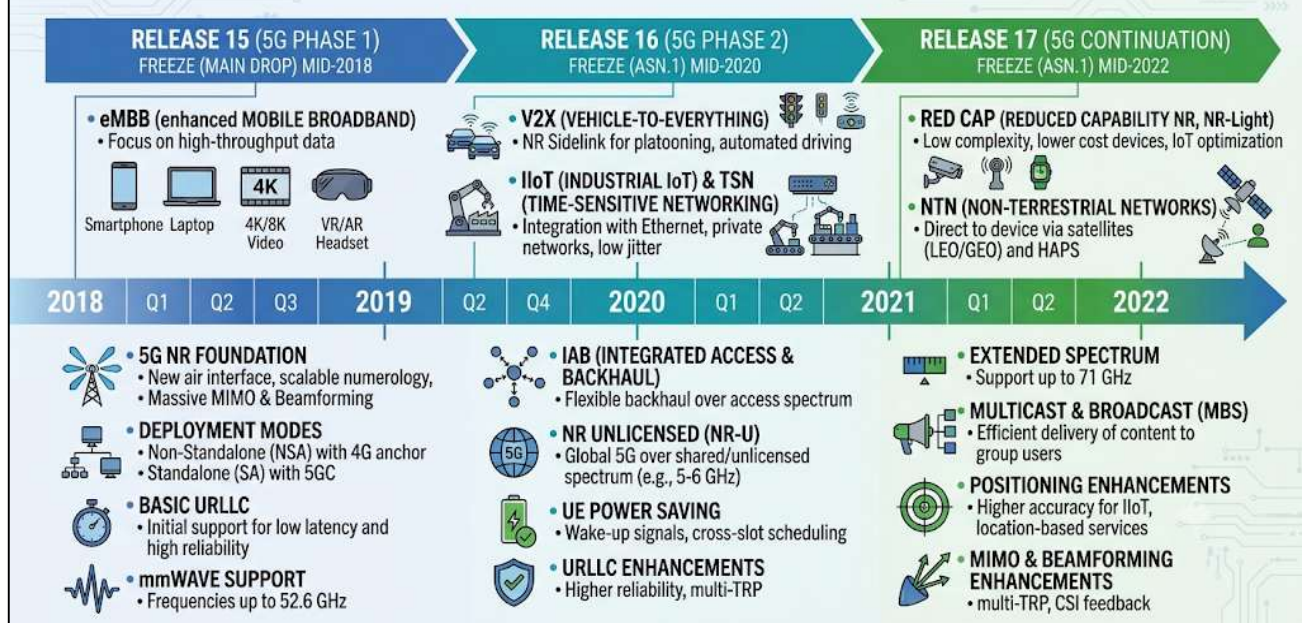


Figure 4.1: Timeline of 3GPP Releases

Release 16

Release 16 significantly expanded 5G capabilities by introducing support for mission-critical applications.

Major improvements include:

- Ultra-Reliable Low-Latency Communication (URLLC)
- Industrial IoT
- Integrated Access and Backhaul (IAB)
- Vehicle-to-Everything (V2X)
- Positioning enhancements
- Network automation

Release 16 is particularly important for intelligent transportation systems and emergency vehicle communication.

Release 17

Release 17 further enhanced 5G performance through:

- Reduced Capability (RedCap) devices

- Improved IoT support
 - Non-Terrestrial Networks (NTN)
 - Enhanced positioning accuracy
 - Satellite communication integration
 - Power-saving improvements
-

Future Releases

Future 3GPP releases continue to improve:

- Artificial Intelligence integration
 - Extended Reality (XR)
 - 6G research
 - Network intelligence
 - Advanced V2X communication
 - Sustainable communication technologies
-

4.2.3 Importance of 3GPP for 5G Deployment

3GPP standards ensure:

- Interoperability
- Vendor compatibility
- Global deployment
- Device compatibility
- Security
- Reliable communication
- Future scalability

These standards enable manufacturers such as Ericsson, Nokia, Samsung, Huawei, Qualcomm, Intel, and others to develop interoperable 5G products.

4.3 ITU IMT-2020 Framework

4.3.1 Introduction

The **International Telecommunication Union (ITU)** is the specialized agency of the United Nations responsible for global telecommunication standards.

For 5G communication systems, ITU introduced the **IMT-2020 (International Mobile Telecommunications-2020)** framework, which defines the minimum performance requirements that a communication technology must satisfy to be officially recognized as a fifth-generation mobile communication system.

Rather than specifying implementation details, the IMT-2020 framework establishes performance objectives covering data rates, latency, reliability, spectrum efficiency, mobility, and connection density.

4.3.2 Performance Requirements

The major IMT-2020 performance targets are summarized below.

Parameter	IMT-2020 Target
Peak Download Speed	20 Gbps
Peak Upload Speed	10 Gbps
User Experienced Data Rate	100 Mbps
End-to-End Latency	1 ms (URLLC Target)
Mobility Support	Up to 500 km/h
Device Density	1 Million Devices/km ²
Reliability	99.999%
Area Traffic Capacity	10 Mbps/m ²

These performance targets enable 5G networks to support diverse applications ranging from high-definition multimedia streaming to autonomous transportation and industrial automation.

4.3.3 Key IMT-2020 Service Categories

The IMT-2020 framework classifies 5G services into three major communication scenarios.

Enhanced Mobile Broadband (eMBB)

Supports:

- Ultra-HD video

- Cloud gaming
 - Virtual reality
 - Augmented reality
 - High-speed Internet
-

Ultra-Reliable Low-Latency Communication (URLLC)

Supports:

- Autonomous vehicles
 - Remote surgery
 - Industrial automation
 - Emergency vehicle communication
 - Smart transportation
-

Massive Machine-Type Communication (mMTC)

Supports:

- Smart cities
 - Internet of Things
 - Smart agriculture
 - Industrial monitoring
 - Environmental sensing
-

4.4 5G New Radio (NR)

5G New Radio (NR) is the global air interface defined by 3GPP.

Major characteristics include:

- Flexible numerology
- Massive MIMO support
- Beamforming
- Low latency

- High spectral efficiency
- Wide bandwidth support

NR operates in both:

- Frequency Range 1 (FR1)
 - Frequency Range 2 (FR2)
-

4.5 Frequency Spectrum

The 5G spectrum is divided into two major frequency ranges.

Frequency Range 1 (FR1)

Frequency:

410 MHz – 7.125 GHz

Characteristics:

- Wide coverage
 - Better penetration
 - Suitable for suburban and rural deployment
-

Frequency Range 2 (FR2)

Frequency:

24.25 GHz – 71 GHz

Characteristics:

- Extremely high bandwidth
 - Multi-gigabit speeds
 - Short communication range
 - Dense urban deployment
-

Comparison

FEATURE	FR1	FR2
FREQUENCY	Sub-7 GHz	mmWave
COVERAGE	Large	Small
DATA RATE	High	Very High
PENETRATION	Better	Lower
DEPLOYMENT	Rural & Urban	Dense Urban

4.6 Quality of Service (QoS)

Quality of Service ensures that different applications receive appropriate network resources according to their communication requirements.

For example:

Application	QoS Requirement
Video Streaming	High Bandwidth
Emergency Communication	Ultra-Low Latency
IoT Sensors	Energy Efficiency
Voice Calls	Low Packet Loss

Network slicing enables operators to allocate dedicated resources to different services.

4.7 Security Standards

5G introduces several improvements over previous generations.

Major security mechanisms include:

- Subscriber authentication
- Mutual authentication
- End-to-end encryption
- Secure identity management
- Network slicing isolation
- Secure communication protocols

These mechanisms protect user privacy while ensuring reliable communication for mission-critical applications.

4.8 Standards Supporting Intelligent Transportation Systems

Modern Intelligent Transportation Systems depend heavily upon standardized communication protocols.

Relevant standards include:

- Cellular Vehicle-to-Everything (C-V2X)
- Vehicle-to-Infrastructure (V2I)
- Vehicle-to-Vehicle (V2V)
- Vehicle-to-Network (V2N)
- Ultra-Reliable Low-Latency Communication (URLLC)

These standards enable connected vehicles to communicate efficiently with traffic management infrastructure.

4.9 Relevance to the Internship Project

The internship project "**Development of Vehicle-to-Infrastructure (V2I)-Based Traffic Signal Preemption for Emergency Vehicles**" is closely aligned with the standardization efforts of 3GPP and the ITU.

The simulation models communication characteristics defined by IMT-2020 and evaluates emergency vehicle preemption under three different traffic management strategies:

- **Baseline (No Preemption):** Conventional fixed-time traffic control without communication.
- **4G LTE Preemption:** Simulated communication latency of approximately **30–50 milliseconds**.
- **5G URLLC Preemption:** Simulated communication latency of **1–5 milliseconds** with ultra-high reliability.

Although the project is implemented within a simulation environment, the communication assumptions are based on internationally recognized 5G performance targets and standards. This ensures that the simulation reflects realistic next-generation transportation scenarios.

4.10 Chapter Summary

This chapter discussed the international standards governing the development and deployment of fifth-generation mobile communication systems. The roles of the **3rd Generation Partnership Project (3GPP)** and the **International Telecommunication Union (ITU)** were presented along with major 3GPP releases, the IMT-2020 performance framework, 5G New Radio (NR), spectrum allocation, Quality of Service, and communication security. These standards establish the technological foundation required for implementing advanced applications such as autonomous transportation, industrial automation, and emergency vehicle traffic signal preemption. With the theoretical background of 5G now complete, the next chapter presents the practical implementation and evaluation of the proposed **Vehicle-to-Infrastructure (V2I)-Based Traffic Signal Preemption for Emergency Vehicles**, including system design, simulation methodology, implementation details, performance evaluation, and comparative analysis.

Chapter 5

Development of Vehicle-to-Infrastructure (V2I)-Based Traffic Signal Preemption for Emergency Vehicles

5.1 Introduction

The rapid growth of urbanization and the continuous increase in the number of vehicles on road networks have resulted in severe traffic congestion in metropolitan cities. Congested roads not only affect the daily commute of the public but also significantly hinder the movement of emergency vehicles such as ambulances, fire engines, and police vehicles. Emergency response vehicles require uninterrupted passage through signalized intersections to reach accident sites, hospitals, or disaster locations within the shortest possible time. However, under conventional traffic management systems, emergency vehicles must follow the same traffic regulations as ordinary vehicles, often becoming trapped at red traffic signals and behind long queues of civilian traffic.

Traditional traffic signal controllers generally operate according to fixed-time signal plans or predefined cyclic schedules. These systems do not possess any mechanism to detect the arrival of an emergency vehicle or dynamically modify signal phases based on real-time traffic conditions. Consequently, emergency vehicles experience unnecessary delays at intersections, reducing the efficiency of emergency response services and potentially affecting public safety.

Recent developments in Intelligent Transportation Systems (ITS) have introduced communication-assisted traffic management techniques that allow vehicles and roadside infrastructure to exchange information in real time. Among these techniques, Vehicle-to-Infrastructure (V2I) communication enables an emergency vehicle to transmit its position, speed, and movement information directly to a traffic signal controller. Based on this information, the traffic controller can temporarily modify signal phases to provide priority to the approaching emergency vehicle while maintaining overall traffic safety.

The evolution of fifth-generation (5G) mobile communication technology has further enhanced the capabilities of V2I communication by providing Ultra-Reliable Low-Latency Communication (URLLC). Compared with previous cellular technologies, 5G URLLC offers significantly lower communication latency and higher reliability, making it highly suitable for safety-critical transportation applications where even small communication delays may affect system performance. The use of direct PC5 sidelink communication further reduces communication delay by allowing direct interaction between the emergency vehicle and the roadside traffic controller without relying on cloud-based routing.

The objective of this internship project is to develop and evaluate a Vehicle-to-Infrastructure (V2I)-based emergency traffic signal preemption system using the SUMO traffic simulator integrated with Python TraCI. The proposed simulation models an ambulance travelling along a two-intersection urban corridor under three different communication scenarios: Baseline (No Communication), 4G LTE Cloud-Routed Preemption, and 5G URLLC Direct PC5 Sidelink Preemption. The performance of these scenarios is compared by analyzing emergency vehicle travel time, stopped delay at intersections, average travel speed, and the impact on surrounding civilian traffic.

Unlike conventional traffic control systems, the proposed implementation models the practical relationship between communication latency, communication range, and traffic signal safety transitions. The study demonstrates that reducing communication latency alone is insufficient unless the preemption request is generated early enough to allow the mandatory traffic signal safety clearance phase to complete before the emergency vehicle reaches the intersection. This relationship forms the central concept of the proposed use case and serves as the basis for evaluating the performance improvements offered by 5G URLLC communication.

The complete implementation has been carried out using SUMO for microscopic traffic simulation, Python TraCI for dynamic traffic signal control, and Streamlit for interactive visualization of simulation results. By combining these software platforms, the project provides a realistic simulation environment for studying emergency vehicle traffic signal preemption under different communication technologies without requiring physical deployment of roadside infrastructure.

The work presented in this chapter focuses entirely on the implementation and evaluation of the proposed Development of Vehicle-to-Infrastructure (V2I)-Based Traffic Signal Preemption for Emergency Vehicles. Beginning with the theoretical background and network modeling, the chapter subsequently discusses the simulation environment, implementation methodology, communication scenarios, performance evaluation, and comparative analysis of Baseline, 4G LTE, and 5G URLLC communication models.

5.2 Theory and Network Modeling

5.2.1 Introduction

Emergency vehicle traffic signal preemption is an important application of Intelligent Transportation Systems (ITS) that enables traffic signals to provide temporary priority to emergency vehicles approaching an intersection. The primary objective of traffic signal preemption is to reduce emergency response time while ensuring the safety of all road users. Unlike conventional traffic management systems that operate according to predetermined timing plans, communication-based preemption systems dynamically modify traffic signal phases after receiving information regarding the location and movement of an emergency vehicle.

The effectiveness of an emergency preemption system largely depends on the communication technology used between the vehicle and the traffic signal controller. Different communication technologies introduce different levels of latency, communication range, and reliability, directly influencing how early the traffic controller can respond to an approaching emergency vehicle. To evaluate these effects, the present project compares three different operational scenarios: Baseline (No Communication), 4G LTE Cloud-Routed Preemption, and 5G URLLC Direct PC5 Sidelink Preemption.

5.2.2 Baseline Traffic Signal Operation

The Baseline scenario represents the conventional traffic management system in which no communication exists between the emergency vehicle and the roadside traffic controller. Under this operating mode, traffic signals continue to follow their predefined cyclic timing irrespective of approaching emergency vehicles.

When an ambulance approaches an intersection operating in Baseline mode, it is treated like any ordinary vehicle. If the traffic signal is red or if civilian traffic occupies the intersection, the ambulance must reduce its speed or stop until the normal signal cycle allows it to proceed. Since the controller receives no advance information regarding the approaching ambulance, no modifications are made to the traffic signal phases.

This conventional approach results in increased travel time, longer stopped delays, and unnecessary interruption of emergency response. The Baseline scenario therefore serves as the reference against which the communication-assisted preemption techniques are evaluated throughout this project.

5.2.3 4G LTE Cloud-Routed Emergency Preemption

The second communication scenario models emergency vehicle preemption using a conventional 4G LTE cellular communication network. In this configuration, communication between the ambulance and the traffic signal controller follows the standard Uu cellular interface.

When the ambulance approaches an intersection, the preemption request is first transmitted to the nearest cellular base station. From there, the communication packet travels through the mobile core network to a centralized cloud server responsible for processing the request. After processing, the command is forwarded to the roadside unit (RSU), which finally delivers the signal preemption request to the traffic signal controller. This multi-hop communication architecture introduces additional transmission and processing delays before the controller receives the emergency request.

In the present simulation, the 4G LTE communication model is represented using a communication latency of approximately **50 milliseconds** with a reliable communication range of **40 metres**. Due to this relatively short trigger distance, the traffic signal controller receives the preemption request only about **2.4 seconds** before the ambulance reaches the intersection. However, the mandatory traffic signal yellow clearance phase requires approximately **4.0 seconds** to safely clear cross traffic. As a result, the ambulance reaches the intersection before the safety transition has completed and is therefore required to stop briefly (for 1.1s) until the green phase becomes available.

Although the communication latency of 4G LTE is relatively low compared to older wireless technologies, the combination of cloud-based routing and shorter communication range limits the overall effectiveness of emergency traffic signal preemption under congested traffic conditions.

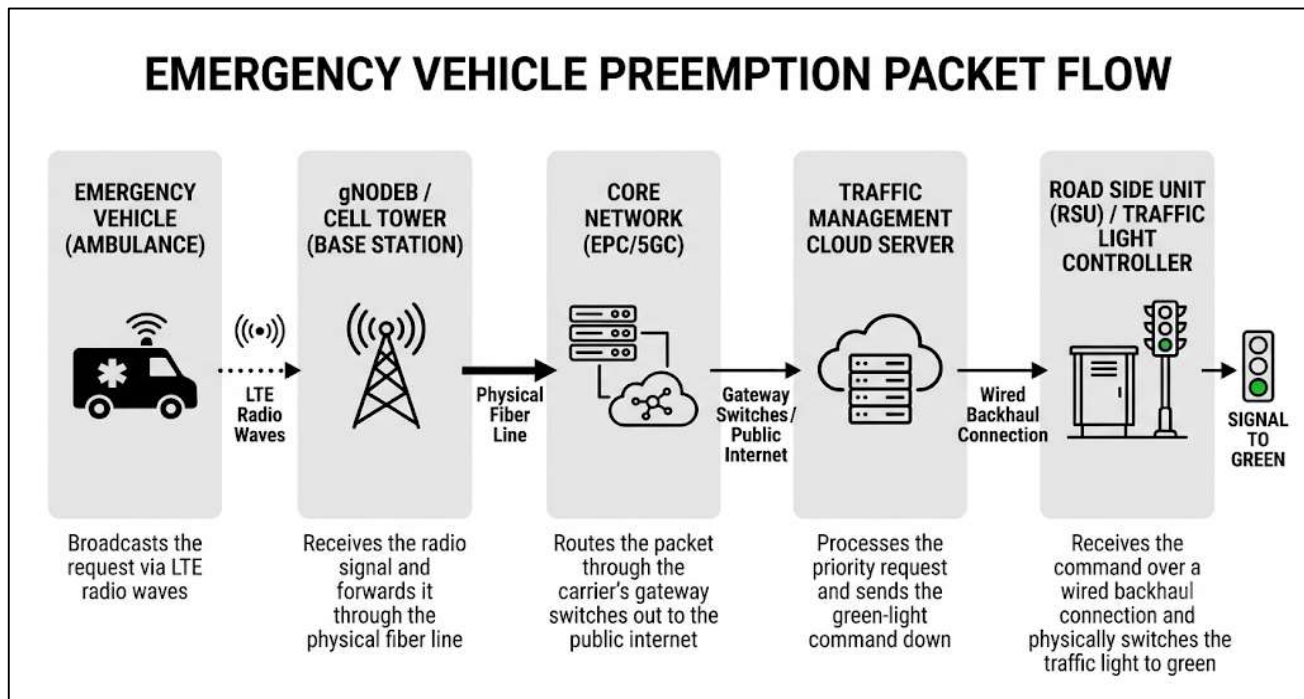


Figure 5.1: 4G LTE Cloud Routing Path showing gNB, Core, Cloud Server, and RSU Interfaces

5.2.4 5G URLLC Direct PC5 Sidelink Preemption

The third communication scenario models emergency traffic signal preemption using **5G Ultra-Reliable Low-Latency Communication (URLLC)** with direct **PC5 sidelink communication**.

Unlike the LTE architecture, the ambulance communicates directly with the roadside traffic controller without routing communication packets through the cellular base station, mobile core network, or cloud server. By eliminating intermediate network components, the

communication delay is significantly reduced while simultaneously increasing communication reliability.

Within the simulation, the 5G communication model is represented using a communication latency of approximately **2 milliseconds** and a reliable communication range of **100 metres**. Because the ambulance establishes communication much earlier than in the LTE scenario, the traffic controller receives the preemption request nearly **6.0 seconds** before the vehicle reaches the intersection, allowing the 4.0s safety yellow phase to clear before the ambulance gets close, so it passes without stopping.

This additional response time enables the traffic controller to complete the mandatory yellow clearance phase well before the ambulance arrives. Consequently, civilian traffic is safely cleared from the intersection, the traffic signal has already transitioned to green, and the ambulance proceeds through the intersection without stopping. The simulation therefore demonstrates that the combination of lower communication latency and greater communication range allows the emergency vehicle to maintain continuous movement while minimizing delays at signalized intersections.

EMERGENCY VEHICLE PREEMPTION: TRADITIONAL 4G VS. 5G SIDELINK (PC5)

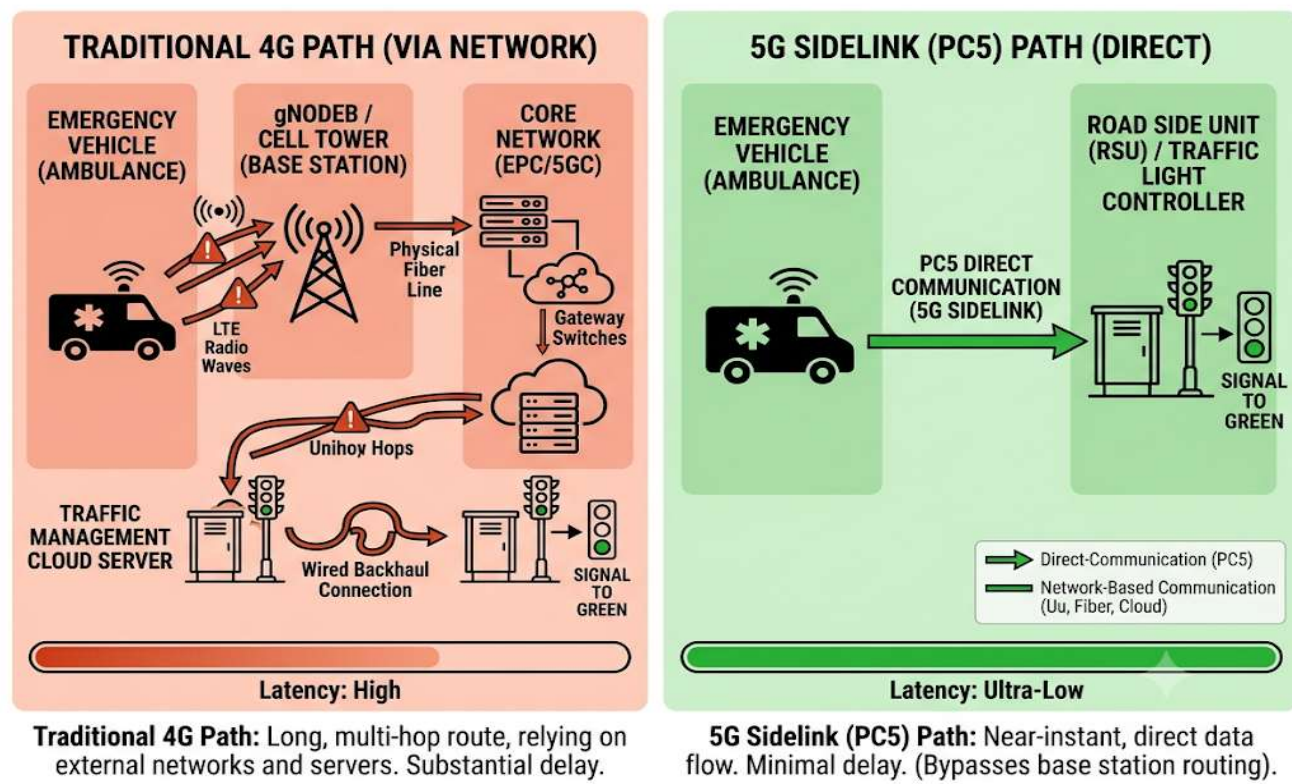


Figure 5.2: 5G PC5 Sidelink Direct Communication Path bypassing Base Station Routing

5.2.5 Traffic Light Safety Transition Dynamics

An important aspect of emergency vehicle traffic signal preemption is the safety transition mechanism followed by traffic signal controllers. Regardless of the communication technology employed, a traffic signal cannot immediately switch from one green phase to another because doing so would create a dangerous conflict between crossing traffic streams.

Instead, whenever a preemption request is received, the controller first initiates a mandatory yellow clearance phase. During this interval, vehicles currently travelling through the active intersection are allowed sufficient time to safely exit before the conflicting traffic movement receives a green indication. In the proposed simulation, this safety clearance interval is modeled as approximately **4.0 seconds**.

The effectiveness of emergency vehicle preemption therefore depends not only on communication latency but also on whether the preemption request is received early enough to complete this mandatory safety transition. If the communication request arrives after the ambulance is already close to the intersection, as in the LTE scenario, the ambulance must still wait for the clearance interval to finish. Conversely, when the request is generated sufficiently early, as in the 5G URLLC scenario, the complete safety transition is finished before the ambulance reaches the stop line, allowing uninterrupted passage through the intersection.

This relationship between communication range, communication latency, and traffic signal safety transition forms the fundamental concept evaluated throughout this project and explains the performance differences observed among the Baseline, 4G LTE, and 5G URLLC communication scenarios.

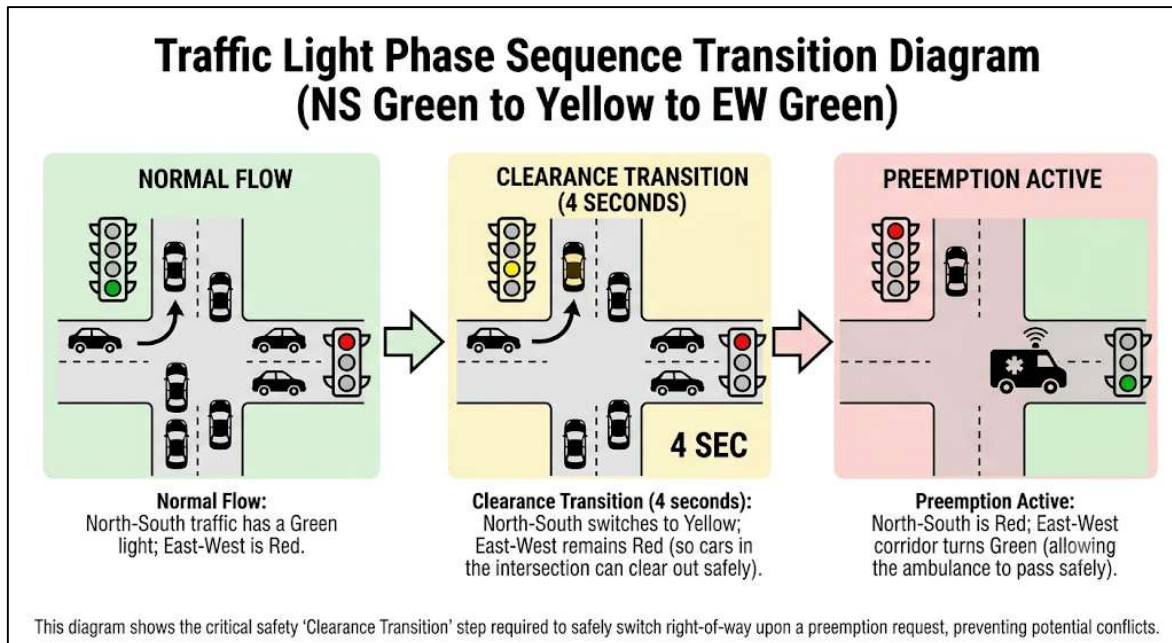


Figure 5.3: Traffic Light Phase Sequence Transition Diagram (NS Green to Yellow to EW Green)

5.3 Implementation Details

5.3.1 Introduction

This section describes the practical implementation of the proposed **Vehicle-to-Infrastructure (V2I)-Based Traffic Signal Preemption for Emergency Vehicles** developed during the internship. The implementation was carried out entirely in a simulation environment using **SUMO (Simulation of Urban MObility)** integrated with **Python TraCI (Traffic Control Interface)**. The objective was to evaluate the performance of different communication technologies for emergency vehicle traffic signal preemption while maintaining identical traffic conditions across all experimental scenarios.

The implementation focuses on comparing three communication models: **Baseline (No Communication)**, **4G LTE Cloud-Routed Preemption**, and **5G URLLC Direct PC5 Sidelink Preemption**. Each communication model follows the same road network, traffic signal configuration, civilian traffic density, and emergency vehicle route. Only the communication mechanism and traffic signal triggering strategy are modified to observe their influence on emergency vehicle movement.

The implementation methodology closely follows the workflow presented in the use case document. The simulation environment models an urban traffic corridor consisting of two signalized intersections, civilian traffic, and a single emergency vehicle travelling along a predefined route. Python continuously communicates with the SUMO simulator using the TraCI interface, monitors the emergency vehicle position, determines the appropriate traffic signal state, and dynamically performs traffic signal preemption whenever required.

5.3.2 Overall System Architecture

The proposed system consists of five primary components working together to perform emergency vehicle traffic signal preemption.

The first component is the **SUMO microscopic traffic simulator**, which models the urban road network, vehicle movement, traffic signal operation, and traffic congestion. SUMO provides a realistic traffic environment where each vehicle behaves independently according to predefined traffic rules.

The second component is the **Emergency Vehicle**, represented in the simulation as an ambulance travelling through the urban corridor. The ambulance follows a predefined route and continuously updates its position during simulation execution.

The third component is the **Python Controller**, which implements the traffic signal preemption algorithm. Python communicates directly with SUMO through the TraCI interface and continuously monitors the location of the emergency vehicle.

The fourth component is the **Traffic Signal Controller**, responsible for changing traffic signal phases whenever a valid preemption request is received. Depending on the communication scenario, the controller either follows the normal traffic signal cycle or temporarily grants signal priority to the approaching ambulance.

The final component is the **Streamlit Dashboard**, which provides visualization of simulation parameters, communication scenarios, and performance results obtained during each experimental run.

The interaction among these components enables real-time monitoring and dynamic control of traffic signals throughout the simulation process.

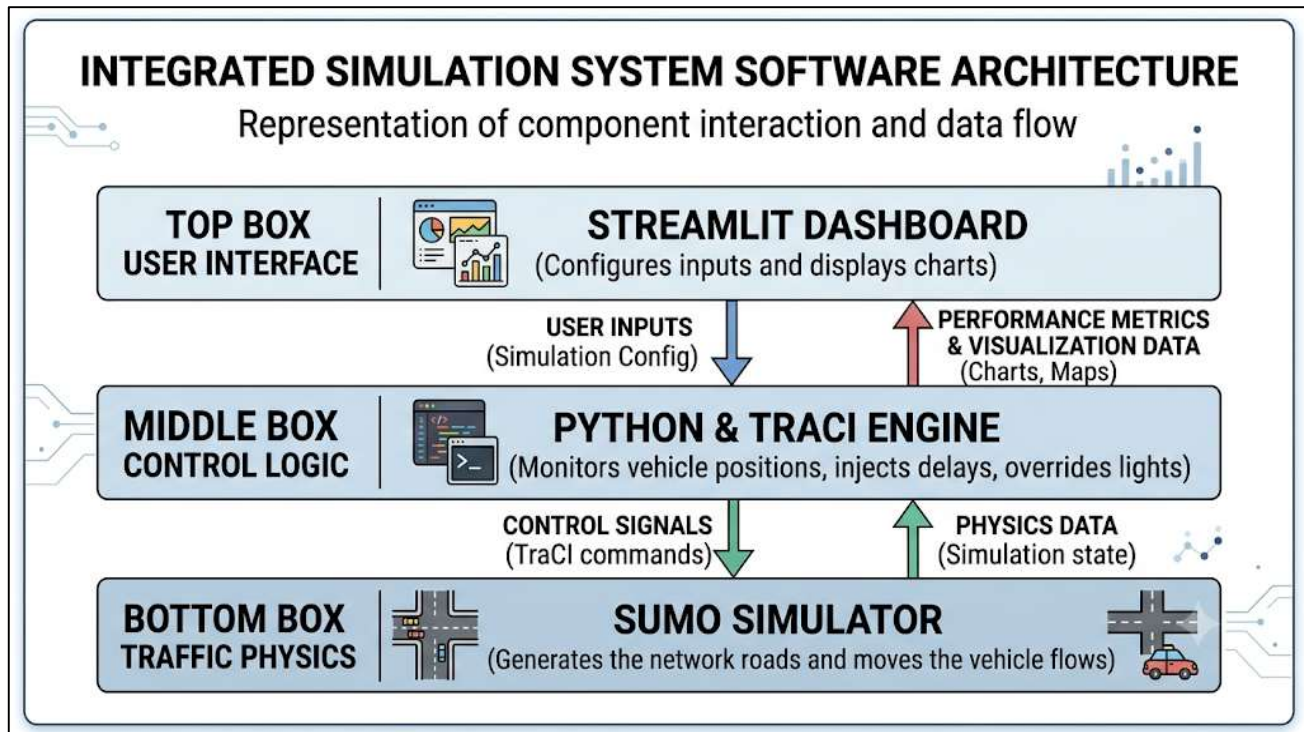


Figure 5.4: Overall System Architecture

5.3.3 SUMO Simulation Environment

The proposed system has been implemented using the **Simulation of Urban MObility (SUMO)**, an open-source microscopic traffic simulator widely used for Intelligent Transportation System (ITS) research.

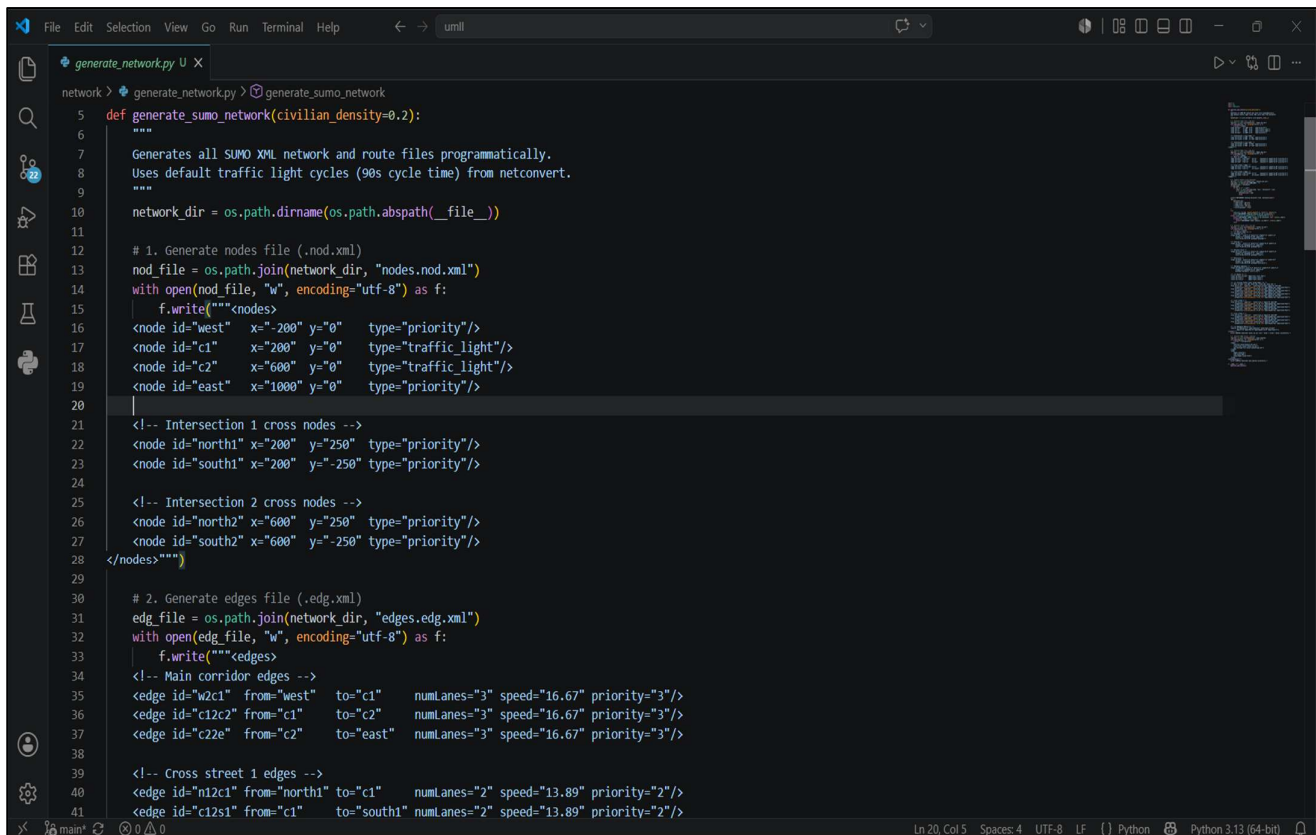
SUMO allows individual modelling of every vehicle, lane, traffic signal, and intersection within a road network. Each vehicle follows realistic acceleration, deceleration, lane-changing, and car-following behaviour based on predefined traffic models. This makes SUMO an appropriate platform for evaluating emergency vehicle movement under varying communication conditions.

For this project, SUMO is responsible for:

- Generating the road network.

- Simulating civilian traffic.
- Simulating emergency vehicle movement.
- Managing traffic signal operation.
- Providing real-time traffic information to Python through TraCI.

The simulator executes the traffic scenario in discrete simulation steps, enabling the Python controller to monitor traffic conditions and modify signal phases dynamically.



```

5 def generate_sumo_network(civilian_density=0.2):
6     """
7     Generates all SUMO XML network and route files programmatically.
8     Uses default traffic light cycles (90s cycle time) from netconvert.
9     """
10    network_dir = os.path.dirname(os.path.abspath(__file__))
11
12    # 1. Generate nodes file (.nod.xml)
13    nod_file = os.path.join(network_dir, "nodes.nod.xml")
14    with open(nod_file, "w", encoding="utf-8") as f:
15        f.write("""<nodes>
16        <node id="west" x="-200" y="0" type="priority"/>
17        <node id="c1" x="200" y="0" type="traffic_light"/>
18        <node id="c2" x="600" y="0" type="traffic_light"/>
19        <node id="east" x="1000" y="0" type="priority"/>
20
21        <!-- Intersection 1 cross nodes -->
22        <node id="north1" x="200" y="250" type="priority"/>
23        <node id="south1" x="200" y="-250" type="priority"/>
24
25        <!-- Intersection 2 cross nodes -->
26        <node id="north2" x="600" y="250" type="priority"/>
27        <node id="south2" x="600" y="-250" type="priority"/>
28    </nodes>""")
29
30    # 2. Generate edges file (.edg.xml)
31    edg_file = os.path.join(network_dir, "edges.edg.xml")
32    with open(edg_file, "w", encoding="utf-8") as f:
33        f.write("""<edges>
34        <!-- Main corridor edges -->
35        <edge id="w2c1" from="west" to="c1" numLanes="3" speed="16.67" priority="3"/>
36        <edge id="c12c2" from="c1" to="c2" numLanes="3" speed="16.67" priority="3"/>
37        <edge id="c22e" from="c2" to="east" numLanes="3" speed="16.67" priority="3"/>
38
39        <!-- Cross street 1 edges -->
40        <edge id="n12c1" from="north1" to="c1" numLanes="2" speed="13.89" priority="2"/>
41        <edge id="c12s1" from="c1" to="south1" numLanes="2" speed="13.89" priority="2"/>

```

Figure 5.5: SUMO Node and Edge XML Code Snippet and Attribute Configurations

5.3.4 Road Network Configuration

A simplified urban corridor was created to evaluate emergency vehicle traffic signal preemption under controlled traffic conditions. According to the project implementation, the corridor contains **two consecutive signalized intersections**, identified as **C1** and **C2**. The total corridor length is approximately **1200 metres**, allowing sufficient distance for emergency vehicle detection and communication before reaching each intersection.

The road network includes:

- Straight urban roadway.
- Two signalized intersections.

- Multiple traffic lanes.
- Civilian vehicle routes.
- Dedicated emergency vehicle route.

The network geometry was generated using SUMO network configuration files and subsequently loaded into the simulation before each experimental run.

The route configuration file routes a continuous flow of civilian vehicles, consisting of passenger cars (50%), buses (15%), delivery trucks (10%), and motorcycles (25%). The ambulance enters the network at $t = 30$ s at the western boundary, traveling east along the main corridor at a speed limit of 60 km/h.

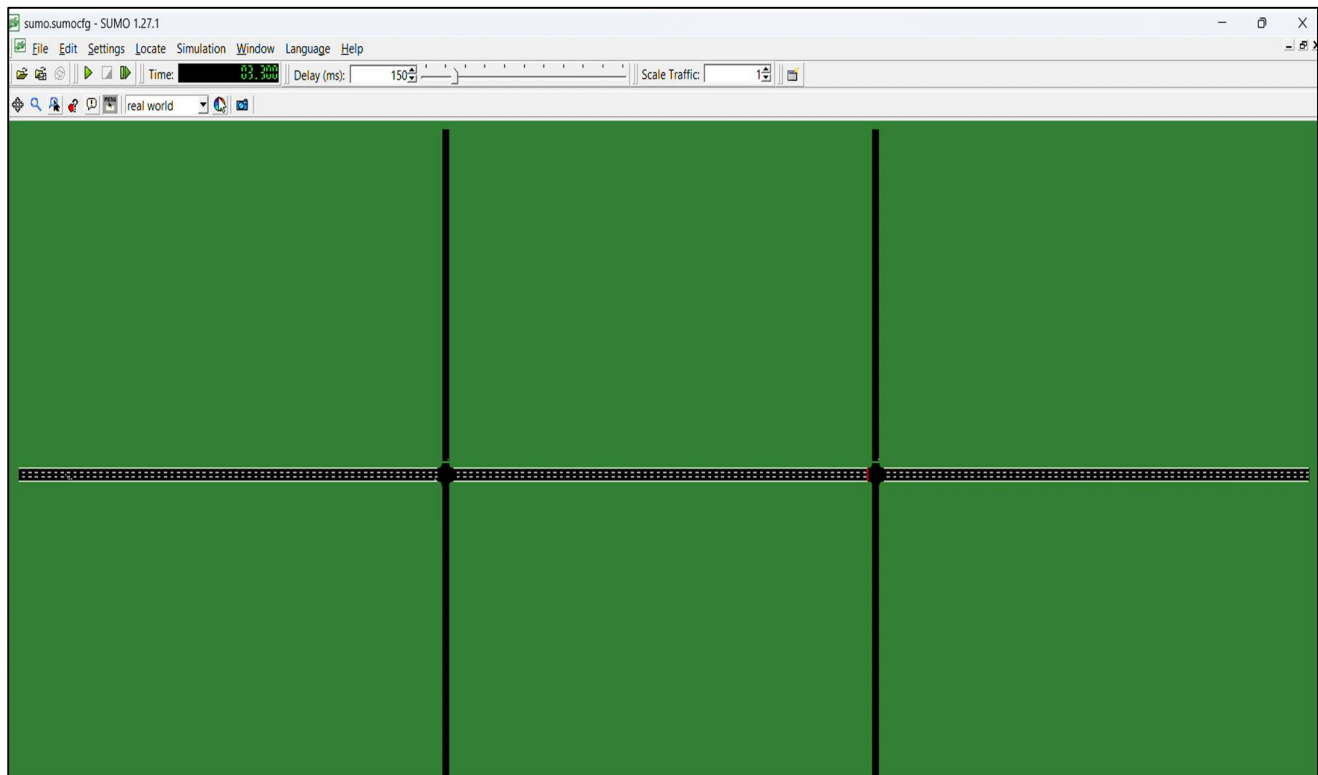


Figure 5.6: SUMO Road Network

5.3.5 Civilian Traffic Generation

To evaluate the influence of communication technology under realistic traffic conditions, civilian traffic was introduced into the simulation corridor using predefined vehicle flows.

The simulated traffic consists of multiple categories of vehicles, including passenger cars, buses, trucks, and motorcycles. These vehicles follow normal traffic regulations without receiving any traffic signal priority. Their movements create realistic traffic queues at

intersections, allowing the emergency vehicle preemption algorithm to be evaluated under practical urban traffic conditions.

The civilian traffic density remains constant throughout all experimental scenarios to ensure a fair comparison between Baseline, 4G LTE, and 5G URLLC communication models.

5.3.6 Emergency Vehicle Configuration

A single ambulance is introduced into the simulation to represent the emergency vehicle. The ambulance follows a predefined route through both signalized intersections and is assigned the highest operational priority within the traffic simulation.

During each simulation step, Python continuously retrieves the ambulance position, speed, and current road segment through the TraCI interface. This information is used to determine the distance between the ambulance and the upcoming intersection.

Unlike civilian vehicles, the ambulance can request temporary traffic signal priority whenever it enters the predefined communication range specified for the selected communication scenario.

5.3.7 TraCI-Based Traffic Signal Controller

The Traffic Control Interface (TraCI) provides a communication bridge between Python and the SUMO simulator.

TraCI enables Python to:

- Read vehicle positions.
- Monitor vehicle speed.
- Detect approaching emergency vehicles.
- Obtain traffic signal status.
- Modify traffic signal phases.
- Restore normal traffic operation after preemption.

During simulation execution, Python performs these operations repeatedly at every simulation step, allowing continuous monitoring of the traffic environment and immediate response whenever an emergency vehicle approaches an intersection.

This real-time interaction between Python and SUMO forms the foundation of the proposed emergency vehicle traffic signal preemption system.

5.3.8 Communication Scenario Implementation

The implementation supports three independent communication scenarios, each representing a different traffic management strategy.

A. Baseline Scenario

In the Baseline configuration, no communication exists between the ambulance and the traffic signal controller. The controller follows its predefined signal timing without considering the presence of the emergency vehicle. Consequently, the ambulance behaves like any ordinary vehicle and must wait whenever the traffic signal is red.

B. 4G LTE Cloud-Routed Scenario

The LTE scenario models communication through the conventional cellular communication path.

The ambulance initiates a preemption request when it is approximately **40 metres** from the intersection. The request passes through the LTE communication path before reaching the traffic signal controller, introducing a communication delay of approximately **50 milliseconds**. Since the request is generated relatively close to the intersection, the yellow clearance phase often remains active when the ambulance arrives, resulting in a brief stop before the signal changes to green.

C. 5G URLLC Direct PC5 Scenario

The 5G URLLC scenario implements direct PC5 sidelink communication between the ambulance and the roadside traffic controller.

The communication request is initiated when the ambulance is approximately **100 metres** from the intersection, providing a significantly larger reaction time than the LTE scenario. Combined with an approximate communication latency of **2 milliseconds**, the traffic controller receives the request early enough to complete the mandatory safety transition before the ambulance reaches the stop line. As a result, the ambulance proceeds through the intersection without stopping.

5.3.9 Traffic Signal Preemption Logic

The traffic signal controller follows a structured sequence of operations to provide priority to the emergency vehicle.

1. The simulation begins with normal traffic signal operation.
2. Python continuously monitors the emergency vehicle.

3. The distance between the ambulance and the next intersection is calculated.
4. The active communication scenario determines the communication delay and trigger distance.
5. If the trigger condition is satisfied, the controller initiates traffic signal preemption.
6. The yellow safety clearance phase is executed.
7. The traffic signal changes to green for the ambulance.
8. After the ambulance clears the intersection, the controller restores the normal traffic signal cycle.

This workflow is repeated independently for every signalized intersection encountered by the emergency vehicle.

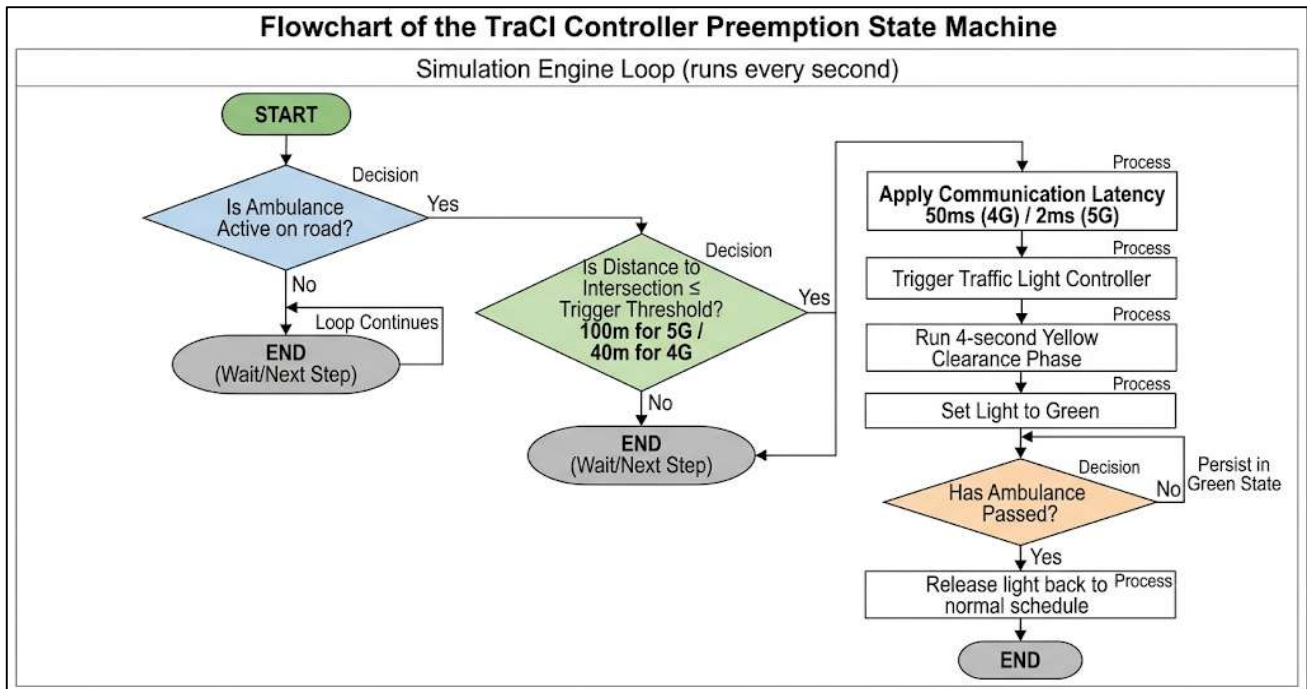


Figure 5.7: Traffic Signal Preemption Flowchart

5.3.10 Visualization Dashboard

A Streamlit-based dashboard was developed to simplify the visualization of simulation results.

The dashboard allows users to:

- Select communication scenarios.
- Start simulations.
- View emergency vehicle statistics.
- Compare communication models.
- Display performance graphs generated using Plotly.

The dashboard provides an intuitive interface for comparing the effects of Baseline, LTE, and 5G URLLC communication on emergency vehicle performance and overall traffic behaviour.

5.3.11 Sectional Summary

This section presented the practical implementation of the proposed Vehicle-to-Infrastructure emergency traffic signal preemption system. The complete implementation was developed using SUMO, Python TraCI, and Streamlit, enabling dynamic traffic signal control under three communication scenarios. The road network, traffic generation, emergency vehicle configuration, communication models, and traffic signal control algorithm were discussed in detail. These implementation components provide the foundation for the experimental evaluation and performance analysis presented in the following section.

5.4 Results and Analysis

5.4.1 Introduction

The effectiveness of the proposed Vehicle-to-Infrastructure (V2I) Emergency Traffic Signal Preemption System was evaluated through a series of simulation experiments conducted using the **Simulation of Urban MObility (SUMO)** integrated with **Python TraCI**. The primary objective of these experiments was to investigate how different communication technologies influence emergency vehicle movement through signalized urban intersections under identical traffic conditions.

Three communication scenarios were evaluated throughout the simulation:

- **Baseline (No Communication)**
- **4G LTE Cloud-Routed Preemption**
- **5G URLLC Direct PC5 Sidelink Preemption**

For each experiment, the road network, civilian traffic density, emergency vehicle route, and traffic signal timing remained unchanged. Only the communication mechanism and corresponding preemption strategy were modified. This ensured that any observed improvement in emergency vehicle performance resulted solely from the communication technology rather than differences in traffic conditions.

The simulation outputs were collected using Python scripts through the TraCI interface and subsequently visualized using Plotly graphs within the Streamlit dashboard. The results provide a comprehensive comparison of emergency vehicle behaviour under conventional traffic management and communication-assisted traffic signal preemption.

5.4.2 Experimental Configuration

To ensure consistency across all experiments, identical simulation parameters were maintained throughout the evaluation process.

The experiments were conducted using the urban corridor developed in SUMO consisting of two signalized intersections (C1 and C2). A single ambulance travelled along a predefined route while civilian traffic generated realistic congestion conditions within the network.

The comparison included the following communication scenarios:

<i>Scenario</i>	<i>Communication Model</i>	<i>Trigger Distance</i>	<i>Approximate Communication Latency</i>
<i>Baseline</i>	No Communication	Not Applicable	Not Applicable
<i>4G LTE</i>	Cloud-Routed Communication	40 m	≈ 50 ms
<i>5G URLLC</i>	Direct PC5 Sidelink	100 m	≈ 2 ms

The communication latency and trigger distance values correspond to those implemented in the proposed use case simulation.

5.4.3 Performance Evaluation Parameters

The performance of each communication scenario was evaluated using several measurable parameters that collectively describe the efficiency of emergency vehicle movement through the traffic network.

Emergency Vehicle Travel Time

Travel time represents the total duration required for the ambulance to travel from its origin to the destination while passing through both signalized intersections. A lower travel time indicates more efficient traffic signal coordination and faster emergency response.

Average Vehicle Speed

Average speed represents the mean travelling speed of the ambulance during the simulation. Continuous movement through intersections generally results in higher average speeds, whereas frequent stopping reduces the overall average speed.

Stopped Delay

Stopped delay measures the amount of time for which the ambulance remains completely stationary because of traffic signal restrictions or queue formation. This metric directly reflects the effectiveness of the preemption strategy.

Traffic Signal Response

Traffic signal response refers to the ability of the controller to modify traffic signal phases before the ambulance reaches the intersection. Earlier response provides more time for clearing civilian traffic and reducing emergency vehicle delay.

5.4.4 Baseline Scenario Analysis

The Baseline scenario represents conventional traffic signal operation without any communication between the ambulance and the roadside traffic controller.

During simulation, traffic signals continued to operate according to their predefined timing plans irrespective of the approaching emergency vehicle. As the ambulance approached each intersection, it was required to obey the current traffic signal state and wait whenever the signal was red or when civilian vehicles occupied the intersection.

Simulation observations indicate that the ambulance experienced the largest stopped delay under this configuration. Since no advance notification was available to the traffic controller, the intersection could not be prepared before the arrival of the emergency vehicle. Consequently, the ambulance encountered traffic queues, reduced its travelling speed, and waited until the normal traffic signal cycle permitted movement.

The Baseline scenario therefore produced the least efficient emergency response among all evaluated communication models and serves as the reference for subsequent comparisons.

Under Baseline operational conditions, the traffic signal controller at Intersection C1 operates strictly on its default fixed cyclic timing scheme (30s Green / 3s Yellow / 30s Red) without any V2I communication capability. As shown in the screenshot, when the emergency ambulance (amb1) approaches the West-to-East corridor (Edge E1), the traffic light remains Red and cross-traffic (North-South) retains priority. Consequently, the ambulance is trapped behind queued background traffic, forcing it to come to a complete stop (0.00 m/s) for 5.20 seconds before the signal naturally transitions to green.



Figure 5.8: Baseline Simulation Output

5.4.5 4G LTE Communication Analysis

The second experiment evaluated emergency vehicle preemption using the simulated 4G LTE communication model.

In this configuration, the ambulance transmitted its preemption request when it entered the 40-metre communication region surrounding the intersection. The request travelled through the cloud-routed LTE communication path before reaching the traffic signal controller.

Although the communication delay was relatively small, the shorter trigger distance meant that the controller received the request only shortly before the ambulance reached the intersection. The controller therefore initiated the mandatory yellow clearance phase, but the ambulance arrived before this safety transition had fully completed.

As a result, the ambulance briefly reduced speed and experienced a short stopped delay before receiving a green signal. Compared with the Baseline scenario, however, both travel time and overall traffic efficiency improved because traffic signal priority was successfully activated.

These observations demonstrate that while LTE communication significantly improves emergency vehicle performance, the available reaction time remains limited because of the shorter communication range and cloud-routed communication process.

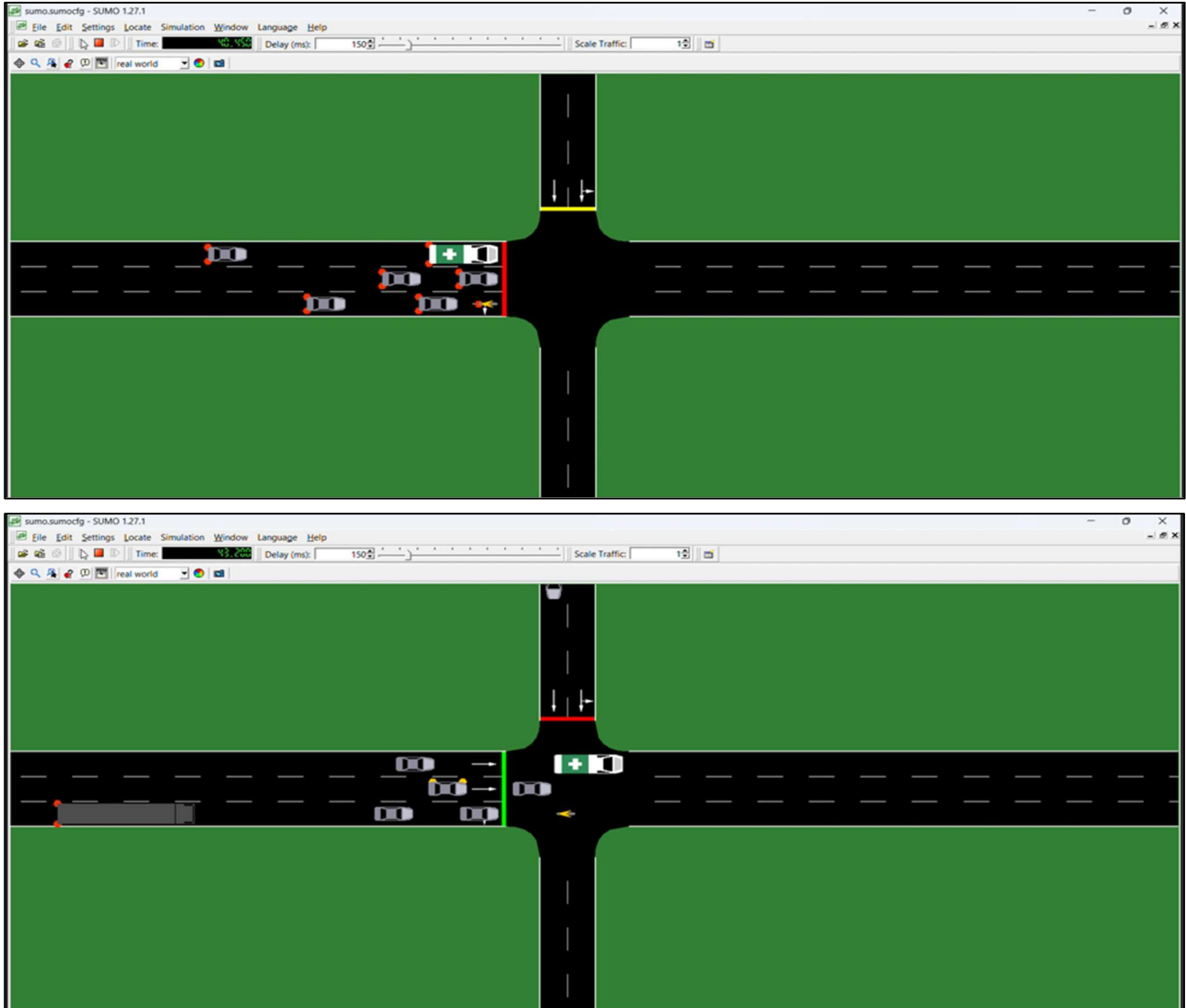


Figure 5.9: 4G LTE Preemption Scenario

In 4G LTE Mode, preemption control messages are transmitted via multi-hop cellular infrastructure (eNodeB → Core Network → Traffic Management Cloud Server → RSU). Due to multi-hop packet routing latency (50.0 ms) and cloud transmission jitter, the preemption trigger distance is constrained to a shorter safety boundary of 40 meters. As illustrated in the screenshot, when the trigger activates at 40m, Intersection C1 initiates a mandatory 3.0s Yellow Clearance Phase for North-South cross traffic. Because the clearance phase is still ongoing when amb1 reaches the stop line, the ambulance is forced to decelerate and briefly stop (1.20 s delay) before receiving full green phase access.

5.4.6 5G URLLC Communication Analysis

The final experiment implemented the proposed **5G URLLC Direct PC5 Sidelink** communication model.

Unlike the LTE scenario, communication between the ambulance and the roadside controller occurred directly through the PC5 interface without intermediate cloud routing. The communication request was initiated approximately 100 metres before the intersection, allowing the controller to receive the request significantly earlier than in the LTE scenario.

Because of this additional preparation time, the traffic controller successfully completed the yellow clearance phase before the ambulance reached the stop line. Civilian vehicles cleared the intersection in advance, enabling the ambulance to continue its journey without stopping.

The simulation clearly demonstrates smoother vehicle movement, uninterrupted passage through both intersections, and the lowest emergency response delay among the three evaluated communication models.

These results confirm that combining ultra-low communication latency with earlier communication initiation substantially improves emergency vehicle mobility in urban traffic environments.

In 5G URLLC Mode, the ambulance utilizes Cellular-V2X (C-V2X) PC5 Sidelink direct communication with sub-1ms radio latency (1.0 ms) to override the roadside traffic controller at intersection C1 directly, bypassing base station routing. This high reliability permits an extended early trigger range of 100 meters. As shown in the screenshot, the 3.0s yellow clearance phase for cross traffic completes while amb1 is still far upstream. By the time the ambulance arrives at intersection C1, the East-West signal has fully transitioned to Green, queue blockages have dissolved, and the vehicle traverses the intersection at full operational speed (15.0 m/s) with 0.00 seconds stopped delay.



Figure 5.10: 5G URLLC Preemption Scenario

5.4.7 V2I Latency and Packet Loss Modeling

The simulation models communication latency using a Gaussian distribution around a configurable base latency. For 4G LTE, we also simulate packet loss. A 10% packet drop rate is modeled, where each drop adds a 200ms retransmission timeout (RTO) before the preemption request is successfully received by the traffic light controller.

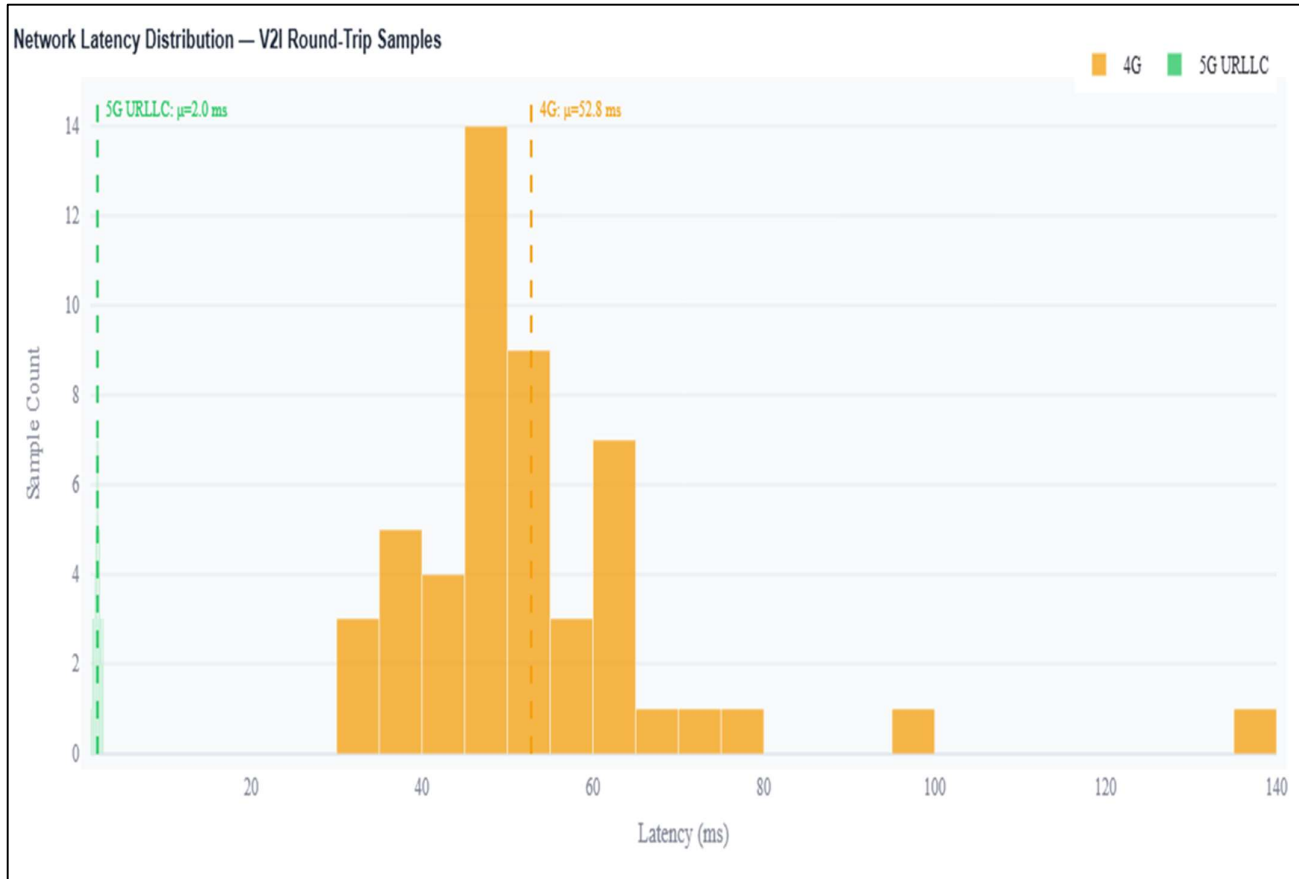


Figure 5.11: Latency Histogram Comparison showing 4G Retransmission Spikes and 5G Stable Profiles

5.4.8 Streamlit Dashboard Frontend Design

A frontend dashboard was built using Streamlit (`app.py`). It provides controls to adjust parameters like civilian traffic density, base latency, and packet loss, and allows launching simulations in either GUI or headless mode. The output displays results using Plotly express charts, showing comparisons of travel times, stopped delays, and speed profiles.

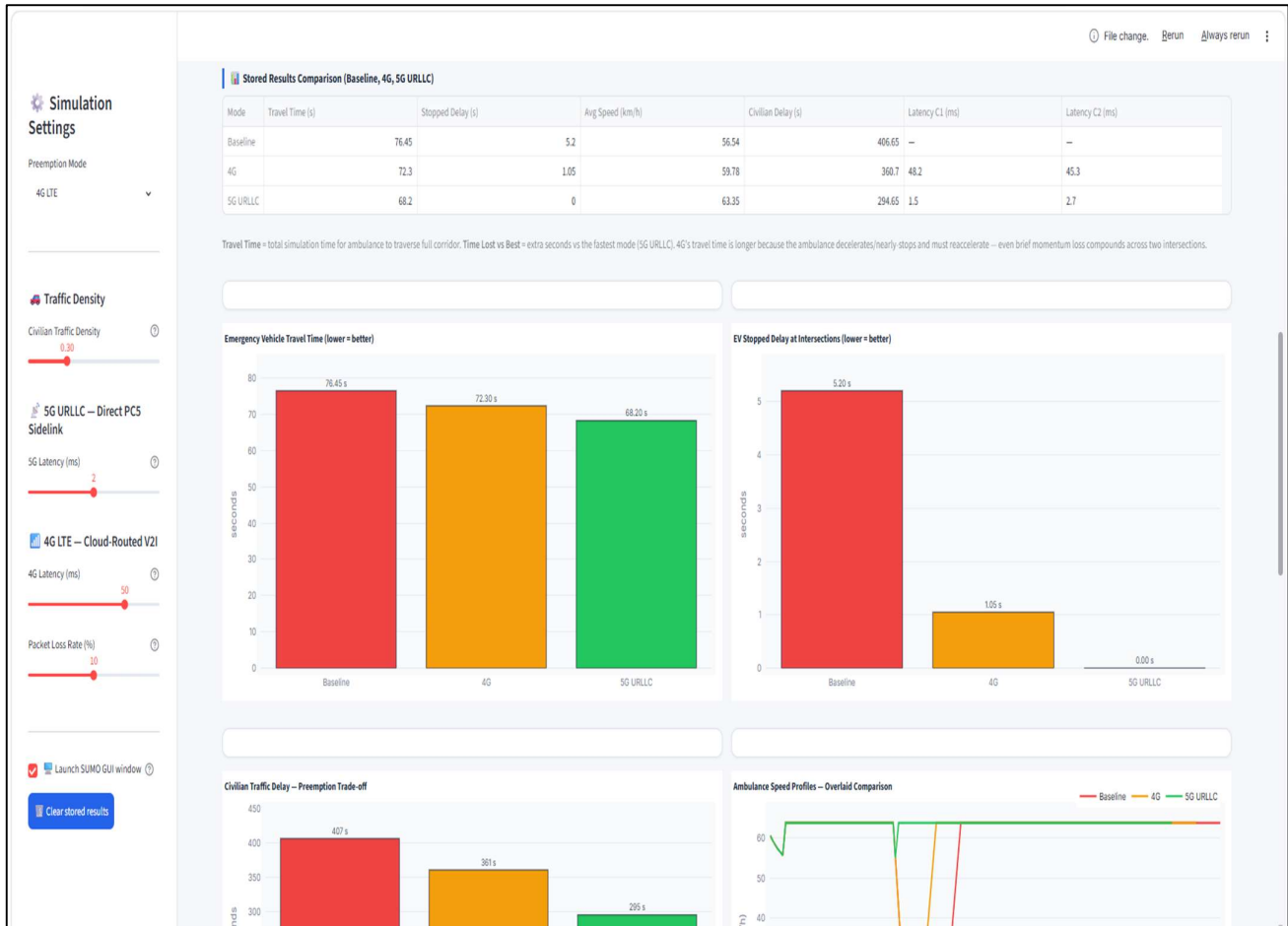


Figure 5.12: Streamlit Dashboard Layout showing Sidebars and Results Visualization Panels

The dashboard frontend uses **Streamlit** to create an interactive interface divided into a configuration sidebar and a main visualization canvas. The left-hand sidebar acts as the control panel, offering dropdown menus and sliders to easily modify simulation settings. Users can toggle the **Launch SUMO GUI window** checkbox to connect the frontend to the traffic simulator, adjust the *Civilian Traffic Density*, and fine-tune network properties. This includes modifying latencies for the **5G URLLC – Direct PC5 Sidelink** configuration, as well as altering latency and *Packet Loss Rate (%)* for the **4G LTE – Cloud-Routed V2I** setup.

On the backend, **Pandas** and **NumPy** instantly process these slider inputs to update the simulation data. The results are displayed in a clean **Stored Results Comparison** table that tracks key metrics like travel times, average speeds, and civilian delays. To make this data simple to interpret, **Plotly** generates responsive, color-coded bar charts for metrics such as *Emergency Vehicle Travel Time* and *EV Stopped Delay*. These charts use a clear red-orange-green color scheme to visually highlight performance improvements, while an overlaid *Ambulance Speed Profiles* line chart maps vehicle acceleration and braking patterns in real time.

5.4.9 Comparative Performance Analysis

The comparative evaluation illustrates the progressive improvement in emergency vehicle performance from the Baseline scenario to LTE preemption and finally to the proposed 5G URLLC implementation.

The simulation was executed under a civilian traffic flow density of 0.3. The comparative metrics are summarized in the table below:

Performance Metric	Baseline (No Preemption)	4G LTE (Cloud-Routed)	5G URLLC (PC5 Sidelink)
V2I Communication Latency (C1)	—	48.2 ms	1.5 ms
Preemption Trigger Distance	—	40 m	100 m
Emergency Vehicle Travel Time	76.45 s	72.30 s	68.20 s
EV Stopped Delay (Intersections)	5.20 s	1.05 s	0.00 s
EV Average Corridor Speed	56.54 km/h	59.78 km/h	63.35 km/h
V2I Communication Latency (C2)	—	45.3 ms	2.7 ms
Civilian Delay	406.65 s	360.7 s	294.65 s

The results demonstrate that under Baseline conditions, the ambulance experiences stopped delays (5.20s) due to red lights and queue blockages. Under 4G LTE, the late trigger (40m) means the yellow clearance phase is still running when the ambulance arrives, causing a brief stop. Under 5G URLLC, the early trigger (100m) allows the yellow clearance to complete before the ambulance arrives, avoiding any stop.

The comparison indicates that increasing the trigger distance while simultaneously reducing communication latency provides the controller with sufficient time to safely transition the traffic signal before the ambulance arrives. Consequently, the emergency vehicle experiences fewer interruptions and maintains smoother movement throughout the corridor.



Figure 5.13: Plotly Chart showing speed profile of the Ambulance under the Three Modes

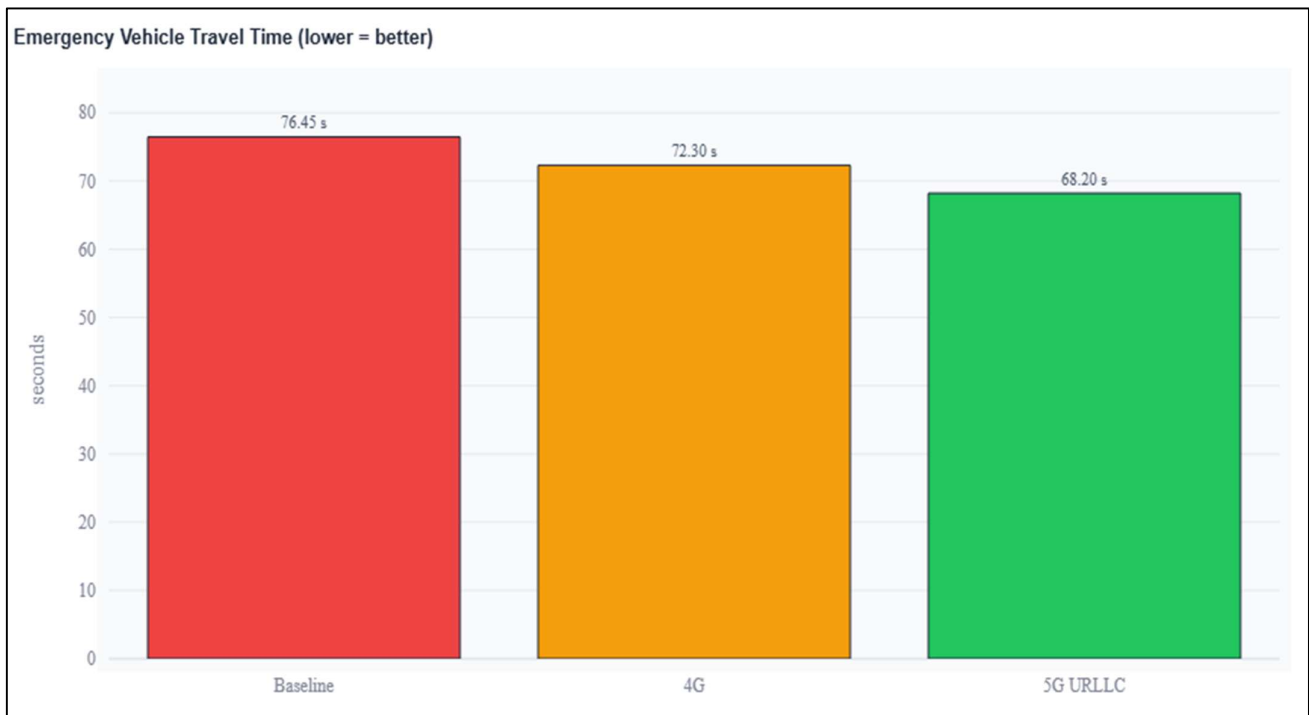


Figure 5.14: Comparison Bar Chart of Total Travel Time Across All Modes

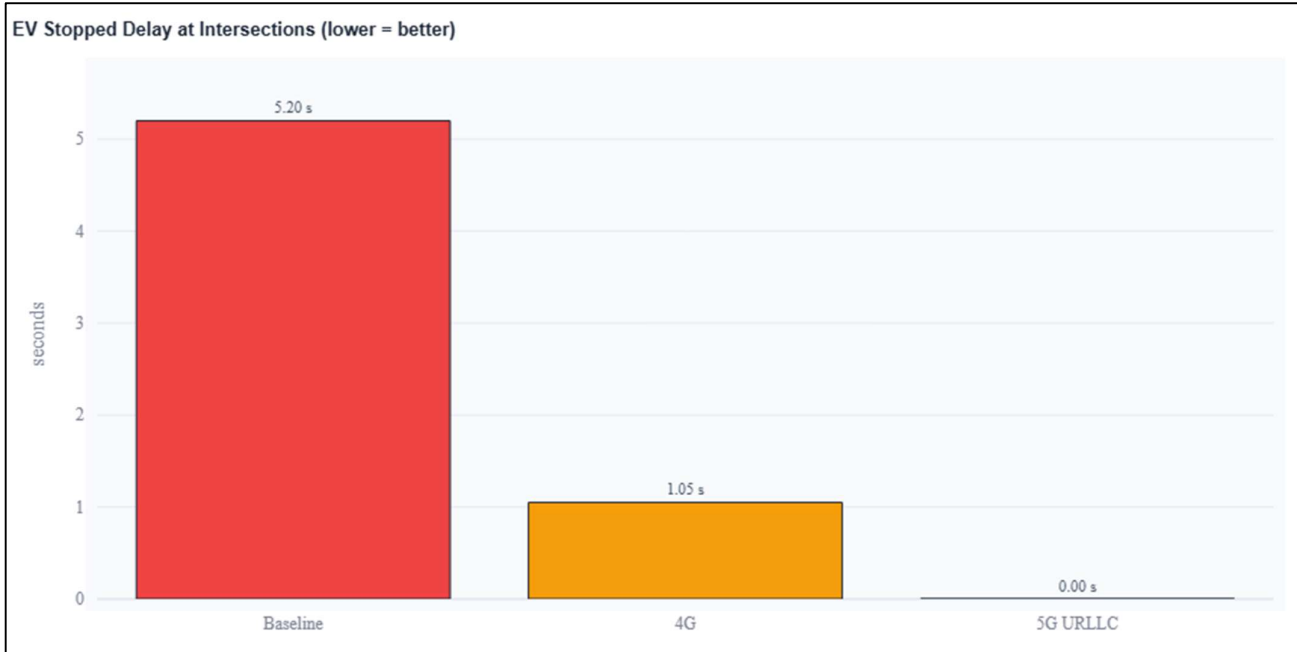


Figure 5.15: Comparison Bar Chart of Stopped Delay Across All Modes

5.4.10 Discussion

The simulation demonstrates that the efficiency of emergency vehicle traffic signal preemption depends on two closely related factors: **communication latency** and **communication initiation distance**.

The Baseline scenario provides no communication support, preventing the traffic controller from responding to the approaching ambulance. The resulting delays arise primarily from fixed traffic signal timing and civilian traffic queues.

The LTE scenario introduces communication-assisted traffic signal control, enabling temporary signal priority. However, because communication begins only when the ambulance is approximately 40 metres from the intersection, the available reaction time is limited. Although the communication delay itself is relatively small, the traffic controller cannot always complete the mandatory yellow clearance phase before the ambulance reaches the intersection.

In contrast, the proposed 5G URLLC implementation combines ultra-low communication latency with a substantially larger communication range of approximately 100 metres. The controller therefore receives the emergency request sufficiently early to complete the safety transition before the ambulance arrives. The intersection is already cleared, allowing uninterrupted passage and eliminating unnecessary stopped delay.

These observations confirm the principal finding of the proposed use case: reducing communication latency alone is insufficient unless the traffic controller also receives the preemption request early enough to safely modify the traffic signal. This relationship between

latency, communication range, and traffic signal clearance represents the central contribution of the present implementation.

5.4.11 Sectional Summary

This section presented the experimental evaluation of the proposed Vehicle-to-Infrastructure Emergency Traffic Signal Preemption System under three communication scenarios: Baseline, 4G LTE, and 5G URLLC. Using identical traffic conditions across all experiments, the simulation demonstrated the influence of communication latency and trigger distance on emergency vehicle performance. The comparative analysis showed that while LTE-based preemption improves emergency response compared with conventional traffic management, the proposed 5G URLLC Direct PC5 communication model provides the most efficient performance by enabling earlier traffic signal preemption, smoother vehicle movement, and reduced emergency response delay. These findings validate the effectiveness of the proposed communication strategy for intelligent transportation systems and emergency traffic management.

Chapter 6

Conclusion and Future Scope

6.1 Conclusion

The primary objective of this internship project was to study the application of **5G Ultra-Reliable Low-Latency Communication (URLLC)** in intelligent transportation systems through the implementation of a **Vehicle-to-Infrastructure (V2I)-Based Traffic Signal Preemption for Emergency Vehicles**. The proposed use case was developed and evaluated using the **Simulation of Urban MObility (SUMO)**, **Python TraCI**, and **Streamlit**, providing a realistic simulation platform for analyzing emergency vehicle movement under different communication technologies.

The study compared three communication scenarios: **Baseline (No Communication)**, **4G LTE Cloud-Routed Preemption**, and **5G URLLC Direct PC5 Sidelink Preemption**. Each scenario was evaluated under identical road network conditions, traffic density, and emergency vehicle route to ensure that the observed performance differences resulted solely from the communication strategy.

The Baseline scenario demonstrated the limitations of conventional fixed-time traffic signal control. Since no communication existed between the ambulance and the traffic signal controller, the emergency vehicle was required to obey normal traffic signals and wait behind civilian traffic whenever a red signal was encountered. This resulted in increased stopped delay, reduced travelling speed, and longer emergency response time.

The 4G LTE communication scenario introduced traffic signal preemption through cloud-routed communication. The simulation showed a noticeable improvement in emergency vehicle movement compared with the Baseline scenario. However, the communication request was initiated only when the ambulance entered the 40-metre trigger region. Although the communication latency remained relatively low, the available reaction time was insufficient to complete the mandatory traffic signal safety transition before the ambulance reached the intersection. Consequently, the ambulance experienced a brief stop while waiting for the yellow clearance phase to complete.

The proposed **5G URLLC Direct PC5 Sidelink** communication model produced the best overall performance among the evaluated scenarios. By increasing the communication trigger distance to approximately 100 metres and reducing communication latency to nearly 2 milliseconds, the traffic controller received the emergency request significantly earlier than in the LTE scenario. This additional preparation time enabled the controller to complete the traffic signal transition before the ambulance arrived, allowing uninterrupted movement through both signalized intersections.

An important observation obtained from the simulation is that **communication latency alone does not completely determine system performance**. The communication trigger distance and the time required for safe traffic signal transition are equally important factors. Even though the LTE communication delay is relatively small, the shorter communication range limits the controller's ability to prepare the intersection in advance. In contrast, the larger communication range combined with ultra-low latency in the 5G URLLC scenario enables effective traffic signal preemption and significantly improves emergency vehicle mobility. This relationship between communication latency, communication range, and traffic signal safety clearance forms the principal contribution of the implemented use case.

The implementation also demonstrates the usefulness of simulation tools for evaluating intelligent transportation systems before real-world deployment. The integration of SUMO, Python TraCI, and Streamlit provides a flexible environment for studying communication-assisted traffic management strategies and comparing different network technologies under controlled conditions.

Overall, the internship successfully achieved its objectives by designing, implementing, and evaluating a Vehicle-to-Infrastructure (V2I)-Based Traffic Signal Preemption for Emergency Vehicles. The results demonstrate the potential benefits of 5G URLLC communication for improving emergency response efficiency and highlight its suitability for future intelligent transportation applications.

6.2 Future Scope

Although the proposed implementation successfully demonstrates the effectiveness of V2I-based emergency traffic signal preemption, several opportunities exist for extending the work in future research and practical deployments.

One possible extension is the development of **multi-intersection corridor coordination**, where multiple traffic signals cooperate to create a continuous green corridor for emergency vehicles travelling through larger urban road networks. Such coordinated traffic management would further reduce emergency response time in densely populated cities.

The present implementation considers a single emergency vehicle. Future work may investigate scenarios involving **multiple emergency vehicles** approaching different intersections simultaneously. Intelligent priority management algorithms would be required to resolve conflicts and allocate traffic signal priority efficiently.

Another possible enhancement involves the use of **real-time traffic information** collected from roadside sensors, connected vehicles, or GPS-based vehicle tracking systems. Incorporating live traffic data would allow traffic signal controllers to make adaptive decisions based on current traffic conditions rather than relying solely on predefined simulation parameters.

The simulation environment may also be expanded by incorporating **larger urban road networks**, higher traffic densities, and more complex intersection layouts. Such studies would provide a better understanding of the scalability and robustness of the proposed traffic signal preemption strategy.

Future implementations may integrate the simulation framework with **real roadside communication hardware**, traffic signal controllers, and vehicle communication devices to validate the proposed methodology under practical operating conditions. Such hardware-based testing would help bridge the gap between simulation studies and real-world intelligent transportation deployments.

Additional research may also investigate the effects of varying communication latency, packet loss, communication reliability, and different traffic conditions to evaluate system performance under a wider range of operating environments.

Finally, the proposed framework may be extended to support broader **Intelligent Transportation System (ITS)** applications, including connected vehicles, adaptive traffic signal control, smart city traffic management, and cooperative Vehicle-to-Infrastructure communication for improving road safety and transportation efficiency. These future directions are consistent with the broader opportunities identified for V2I communication and emergency traffic management in the use case document.

6.3 Internship Learning Outcomes

The internship provided valuable practical experience in the field of fifth-generation communication systems and intelligent transportation technologies. Through the implementation of the proposed use case, practical knowledge was gained in traffic simulation, communication system modelling, and software-based traffic signal control.

The internship contributed to the development of technical skills in:

- SUMO (Simulation of Urban MObility)
- Python programming
- TraCI (Traffic Control Interface)
- Streamlit dashboard development
- Plotly and Matplotlib data visualization
- Vehicle-to-Infrastructure (V2I) communication concepts
- 5G URLLC communication principles
- Traffic signal preemption algorithms
- Performance evaluation and simulation analysis

In addition to technical knowledge, the internship enhanced problem-solving ability, analytical thinking, simulation-based research methodology, and technical documentation skills. These experiences provide a strong foundation for future work in wireless communication, intelligent transportation systems, and smart city technologies.

6.4 Final Remarks

The successful implementation of the **Vehicle-to-Infrastructure (V2I)-Based Traffic Signal Preemption for Emergency Vehicles** demonstrates how communication-assisted traffic management can improve emergency vehicle mobility in urban transportation networks. The comparative analysis of Baseline, 4G LTE, and 5G URLLC communication scenarios highlights the importance of both communication latency and communication range in achieving effective traffic signal preemption.

The study confirms that 5G URLLC communication enables earlier traffic signal activation, smoother emergency vehicle movement, and improved traffic management compared with conventional traffic control methods. Although the present work was conducted in a simulation environment, the implementation provides a meaningful foundation for future research and practical deployment of intelligent transportation systems that support safer, faster, and more efficient emergency response services.

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